

August-November 2024

CS616: Statistical Pattern Recognition

CS612: Statistical Pattern Recognition Laboratory

Programming Assignment 4 Report

Submitted by Group 03

Pawan Ram Mali (PhD)

Department of CSE
Indian Institute of
Technology
Dharwad, India
cs24dp011@iitdh.ac.in

Abhishek Sharma (MS)

Department of CSE
Indian Institute of
Technology
Dharwad, India
cs24ms003@iitdh.ac.in

Sachin Maurya (MS)

Department of CSE
Indian Institute of
Technology
Dharwad, India
cs24ms002@iitdh.ac.in

Contents

I. Introduction	5
A. Overview of the Assignment and Objectives.....	5
B. Datasets Description	5
Dataset 1: 2-dimensional artificial data	5
Dataset 2: 3-class scene image dataset.....	5
II. Dataset 1: 2-Dimensional Artificial Data.....	6
A. Methodology	6
1. Bayes Classifier Using KNN Density Estimation	6
2. Fisher Linear Discriminant Analysis (FDA).....	6
3. Perceptron-Based Classifier.....	6
4. Logistic Regression Classifier	6
5. SVM-Based Classifier	6
B. Results and Analysis	7
1. Classifier Performance Metrics.....	28
2. KNN-Based Bayes Classifier Analysis.....	45
3. FDA Analysis	45
4. Perceptron Analysis	46
5. Logistic Regression Analysis.....	46
6. SVM Analysis.....	46
C. Comparative Analysis	46
1. Decision Region Plots	46
2. Special Comparisons.....	46
III. Dataset 2: 3-Class Scene Image Dataset.....	47
A. Methodology	47
1. Bayes Classifier Using GMM	47
2. Bayes Classifier Using KNN	47
3. FDA Implementation	47
4. Perceptron Implementation.....	47
5. Logistic Regression Implementation	47
6. SVM Implementation	47
B. Results and Analysis	48

1. Classifier Performance Metrics.....	Error! Bookmark not defined.
2. GMM Analysis	Error! Bookmark not defined.
3. KNN Analysis	Error! Bookmark not defined.
4. FDA Performance	Error! Bookmark not defined.
5. Perceptron and Logistic Regression Results.....	Error! Bookmark not defined.
6. SVM Results	Error! Bookmark not defined.
C. Dimensionality Reduction Analysis	71
1. PCA Analysis	71
2. FDA Analysis.....	71
IV. Overall Comparative Analysis	72
A. Cross-Assignment Comparison	72
V. Conclusion	73
A. Key Findings	73
B. Best Performing Methods	73
C. Analysis Insights.....	73
VI. References	74

List of Figures

Figure 1 GMM based Classifier Decision Regions for different number of components ..	7
Figure 2 GMM Cluster centers for different GMM components	8
Figure 3 KNN Decision Regions	9
Figure 4 Original Space & Projected Space in FDA.....	10
Figure 5 Perceptron Decision Regions	11
Figure 6 Logistic Regression Decision Regions	11
Figure 7 SVM Decision Regions	12
Figure 8 SVM Decision Regions	13
Figure 9 GMM Based Classifier Decision Regions for Components 1 to 8	15
Figure 10 GMM Based Classifier Decision Regions for Components 16	16
Figure 11 GMM Cluster centers for GMM Components 1 to 2	17
Figure 12 GMM Cluster centers for GMM Components 4 to 8	18
Figure 13 GMM Cluster centers for GMM Components 16.....	19
Figure 14 FDA 1D Projection	19
Figure 15 Perceptron Decision Boundary	20
Figure 16 Logistic Regression Decision Boundary	20
Figure 17 SVM Linear Decision Boundary	21
Figure 18 SVM Polynomial Decision Boundary for Degree = 2	22
Figure 19 SVM Polynomial Decision Boundary for Degree = 3	23
Figure 20 SVM Polynomial Decision Boundary for Degree = 4	24
Figure 21 SVM RBF Decision Boundary (gamma =0.01)	25
Figure 22 SVM RBF Decision Boundary (gamma =0.1)	26
Figure 23 SVM RBF Decision Boundary (gamma =1)	27
Figure 24 Variation of Eigenvalues.....	48

I. Introduction

A. Overview of the Assignment and Objectives

This assignment aims to apply statistical pattern recognition techniques to two datasets: a 2D artificial dataset and a 3-class scene image dataset. The primary objectives are:

1. Implementation of classifiers, including Bayes classifier, KNN-based density estimation, FDA, perceptron, logistic regression, and SVM with various kernels.
2. Evaluation of classifier performance using metrics such as accuracy, precision, recall, and F-measure.
3. Visualization and analysis of decision boundaries, dimensionality reduction techniques, and cross-method comparisons.

B. Datasets Description

Dataset 1: 2-dimensional artificial data

- Contains two subsets:
 - **Linearly separable dataset:** Suitable for evaluating classifiers with linear decision boundaries.
 - **Nonlinearly separable dataset:** Tests classifiers' abilities to handle non-linear relationships.

Dataset 2: 3-class scene image dataset

- Represented as 32-dimensional Bag-of-Visual-Words (BoVW) features derived from images.
- Requires dimensionality reduction (e.g., PCA) to manage high dimensionality.

II. Dataset 1: 2-Dimensional Artificial Data

A. Methodology

1. Bayes Classifier Using KNN Density Estimation

- Experimented with different K values (e.g., $K=1, 3, 5, 7$).
- Implementation details:
 - Density estimation was performed using KNN for both datasets.
 - Decision boundaries were visualized to analyze the impact of K on classification performance.

2. Fisher Linear Discriminant Analysis (FDA)

- Implemented FDA to reduce data dimensionality and trained Bayes classifiers:
 - Using unimodal Gaussian densities.
 - Using GMMs with varying numbers of mixtures.

3. Perceptron-Based Classifier

- Implementation:
 - Trained perceptron using a standard update rule on linearly and nonlinearly separable datasets.
- Training focused on convergence and robustness against non-linearity.

4. Logistic Regression Classifier

- Applied a gradient-descent approach to optimize parameters.
- Regularization techniques were explored to prevent overfitting.

5. SVM-Based Classifier

- Implemented SVM using:
 - Linear kernel.
 - Polynomial kernel with varying degrees.
 - Gaussian/RBF kernel with different bandwidth parameters.
- Parameters were tuned to optimize performance for both linear and non-linear datasets.

B. Plot Visualizations for Linearly Separable Dataset

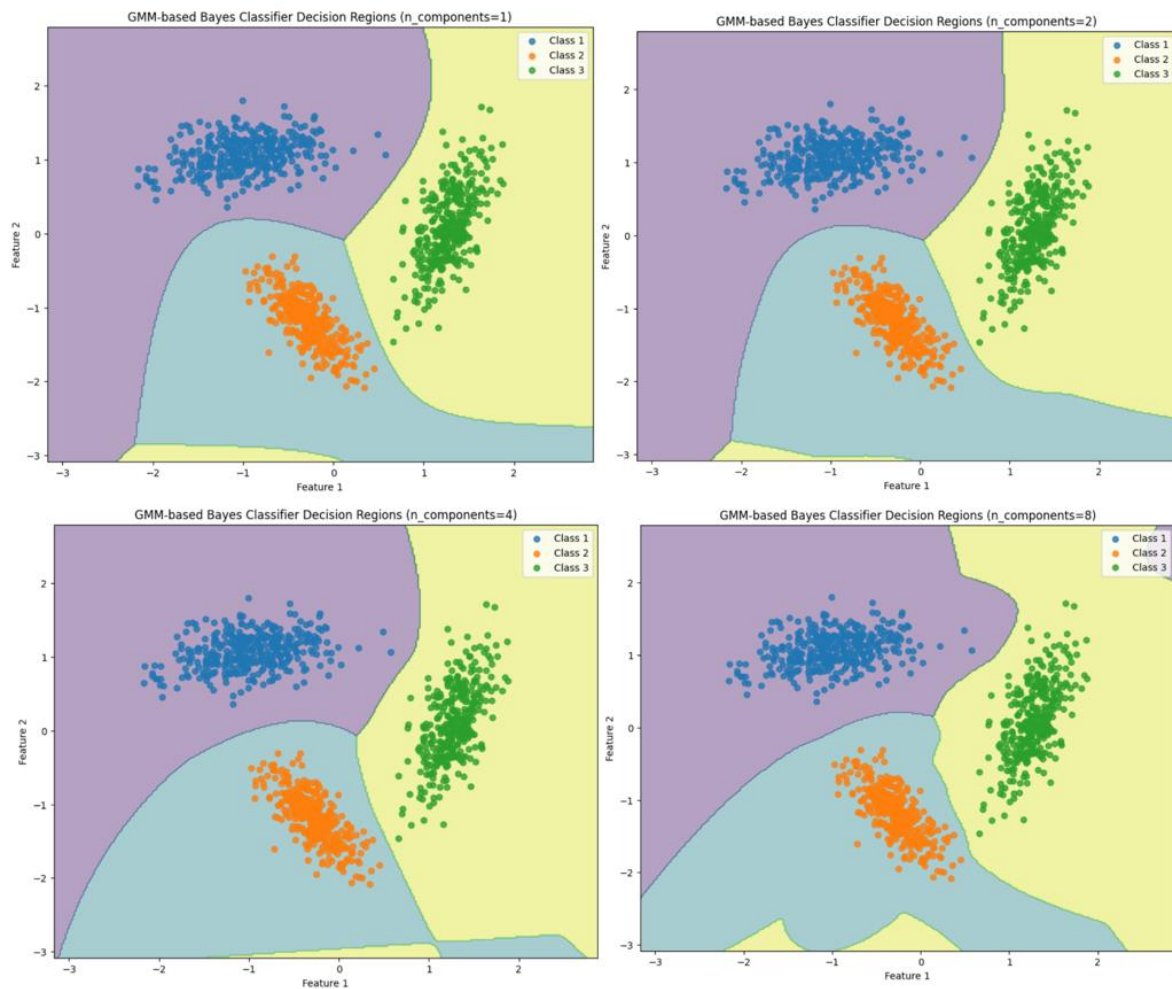


Figure 1 GMM based Classifier Decision Regions for different number of components

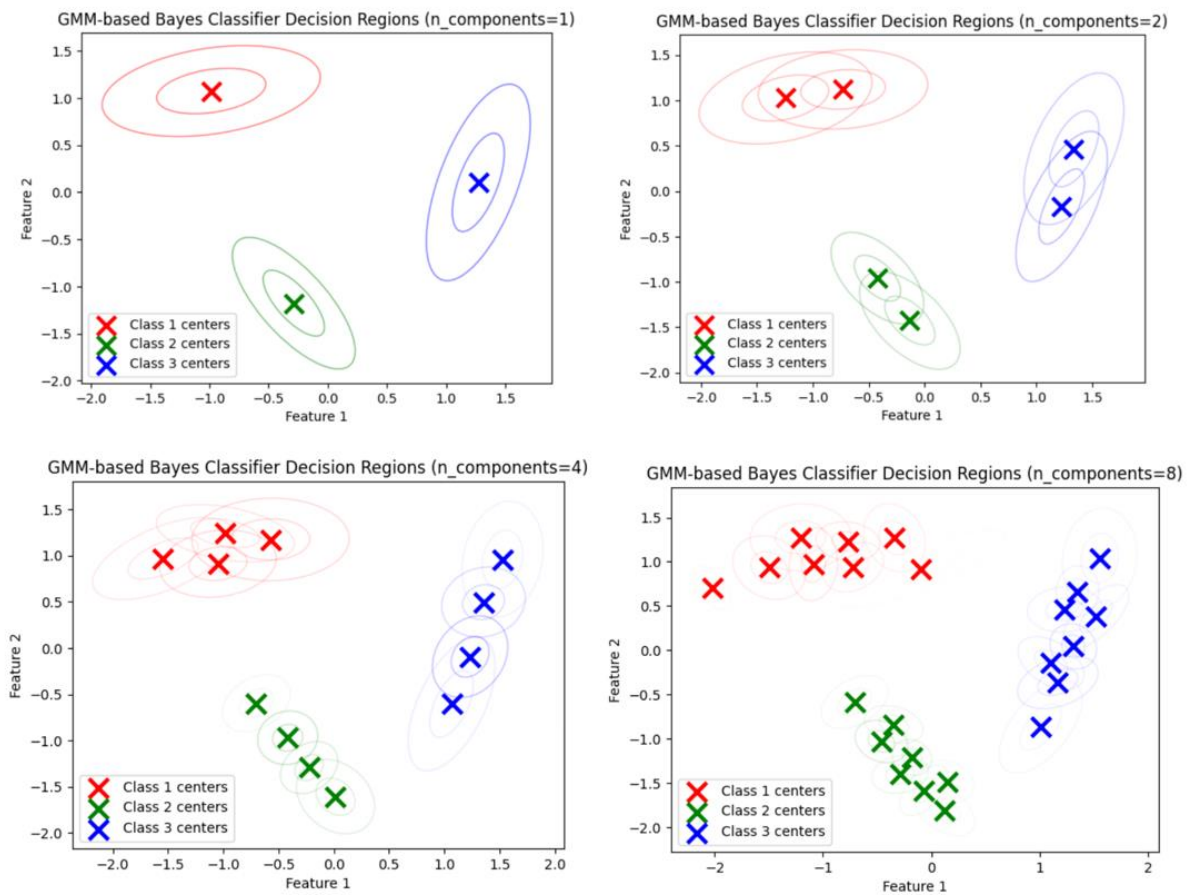


Figure 2 GMM Cluster centers for different GMM components

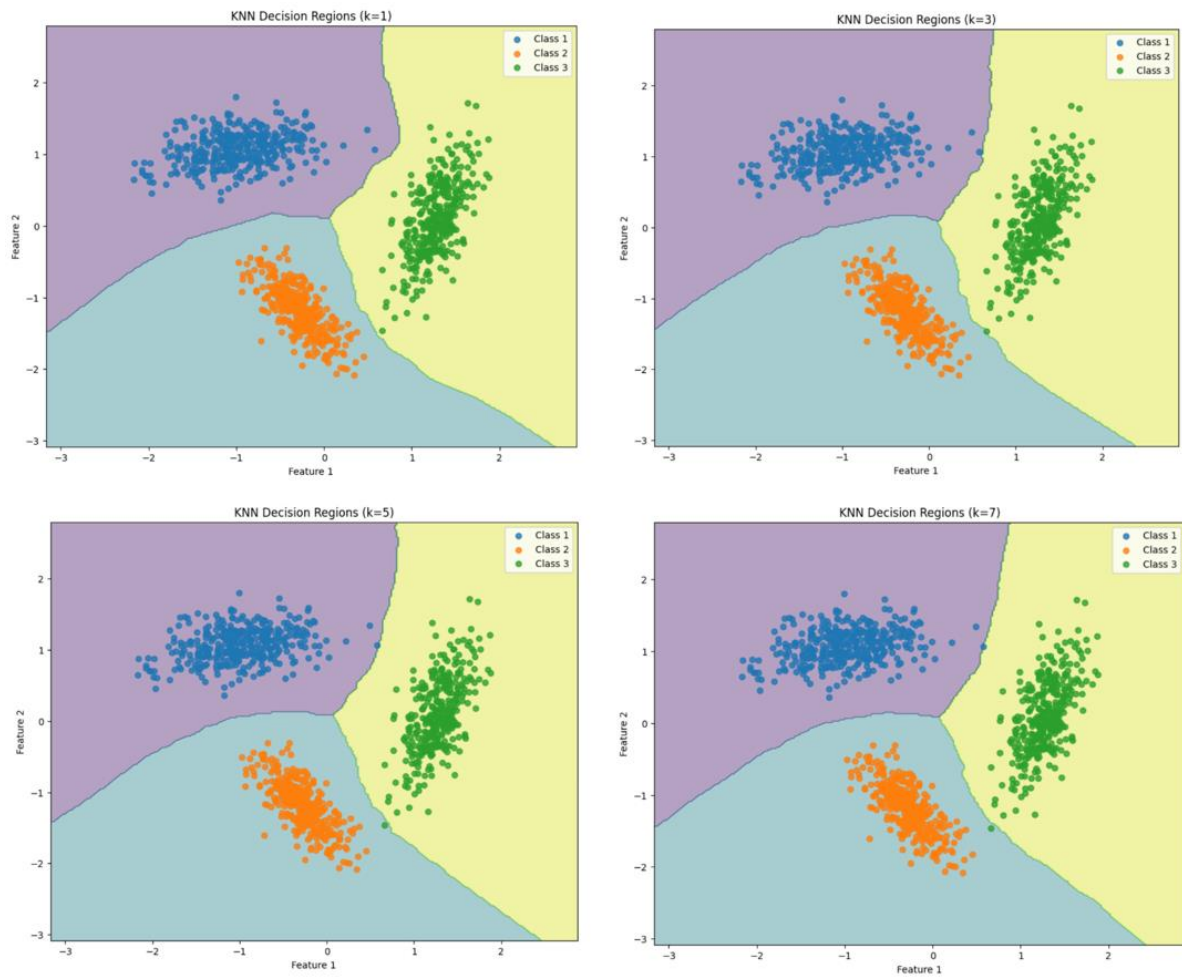


Figure 3 KNN Decision Regions

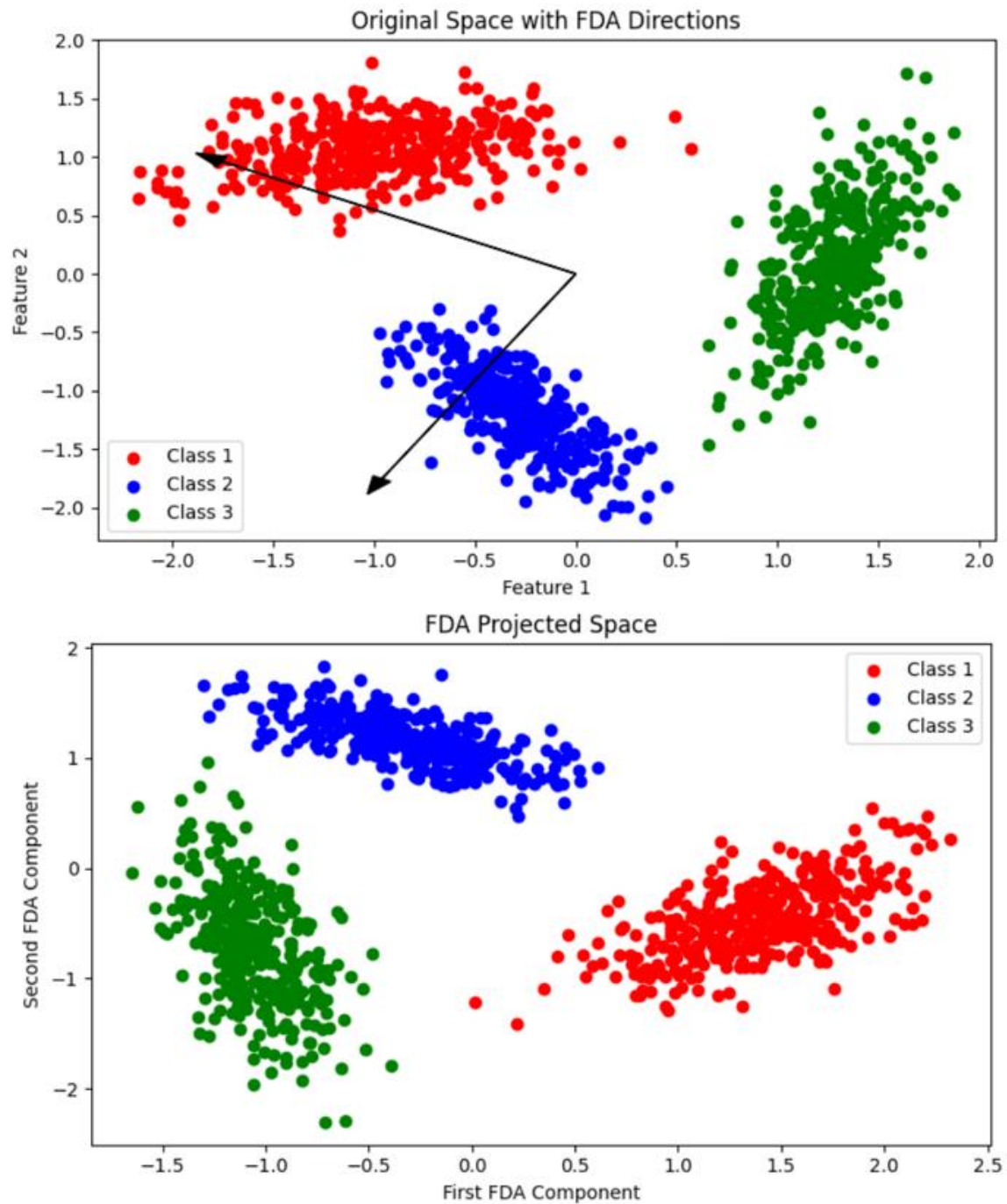


Figure 4 Original Space & Projected Space in FDA

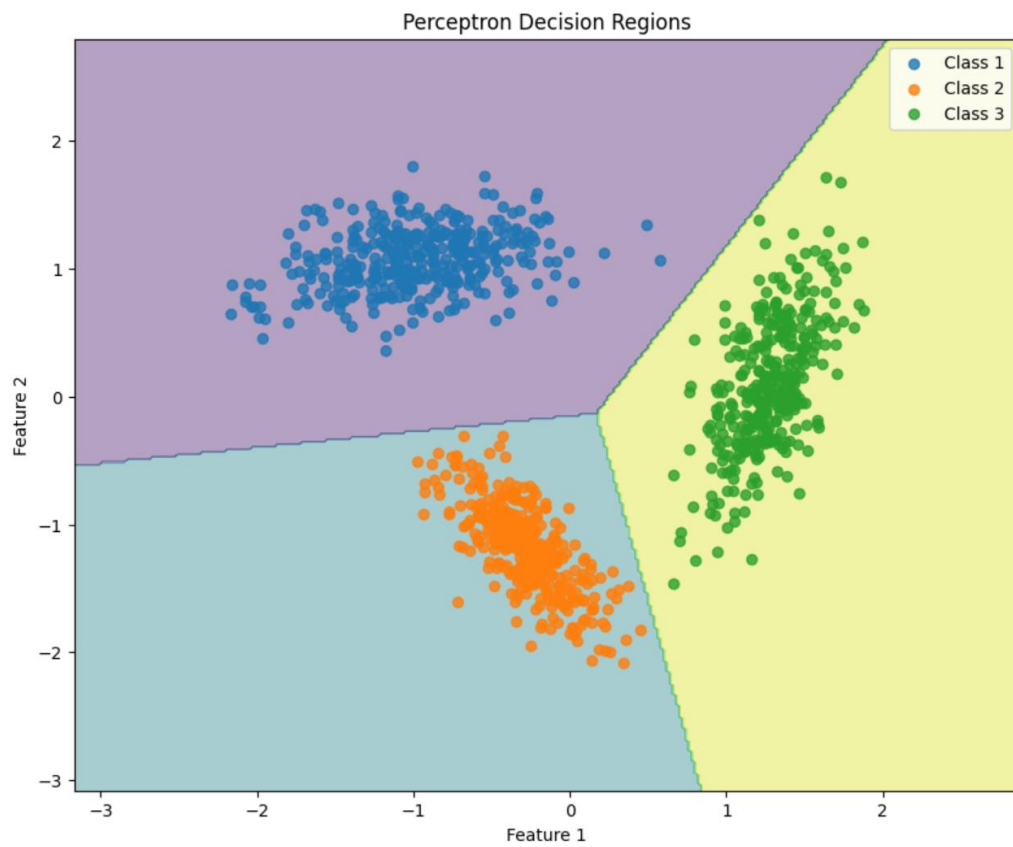


Figure 5 Perceptron Decision Regions

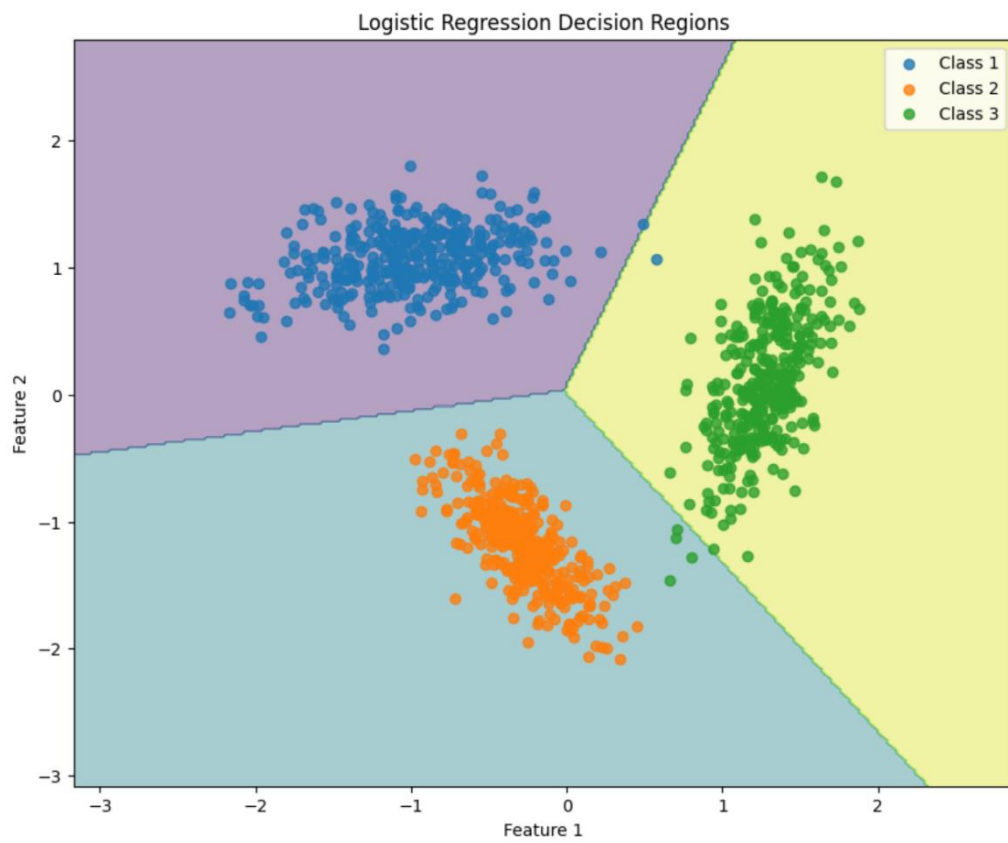


Figure 6 Logistic Regression Decision Regions

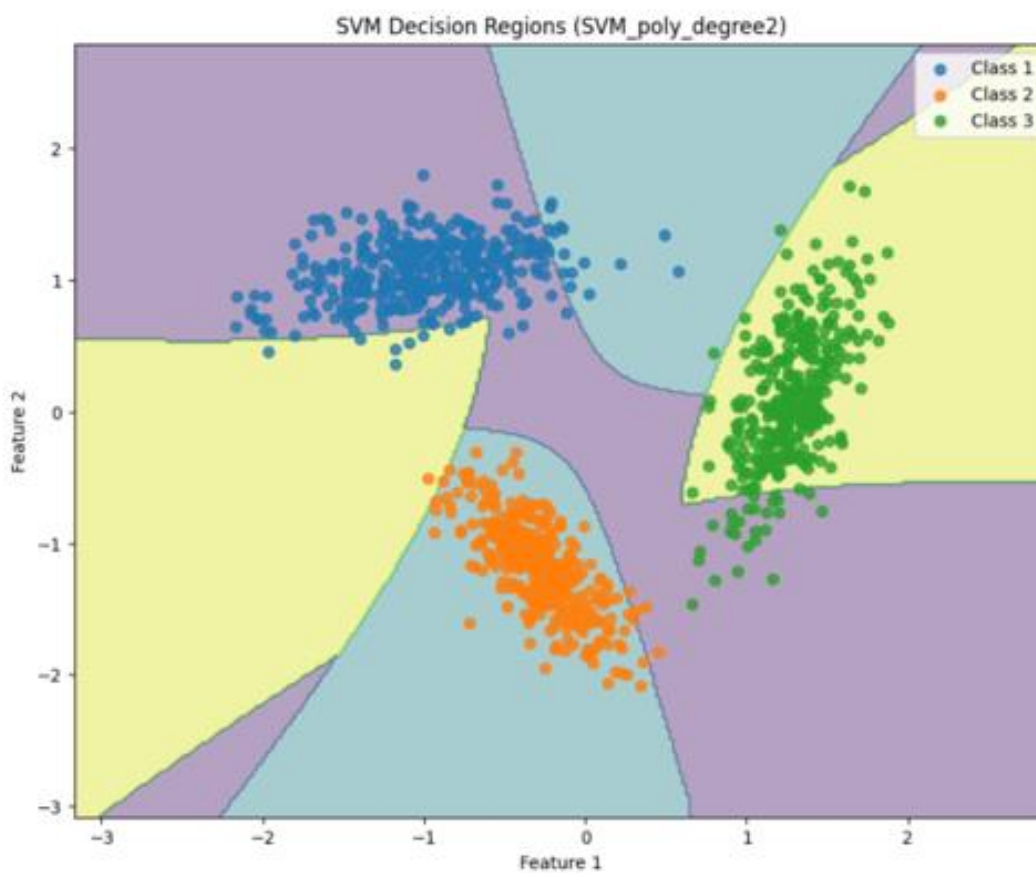
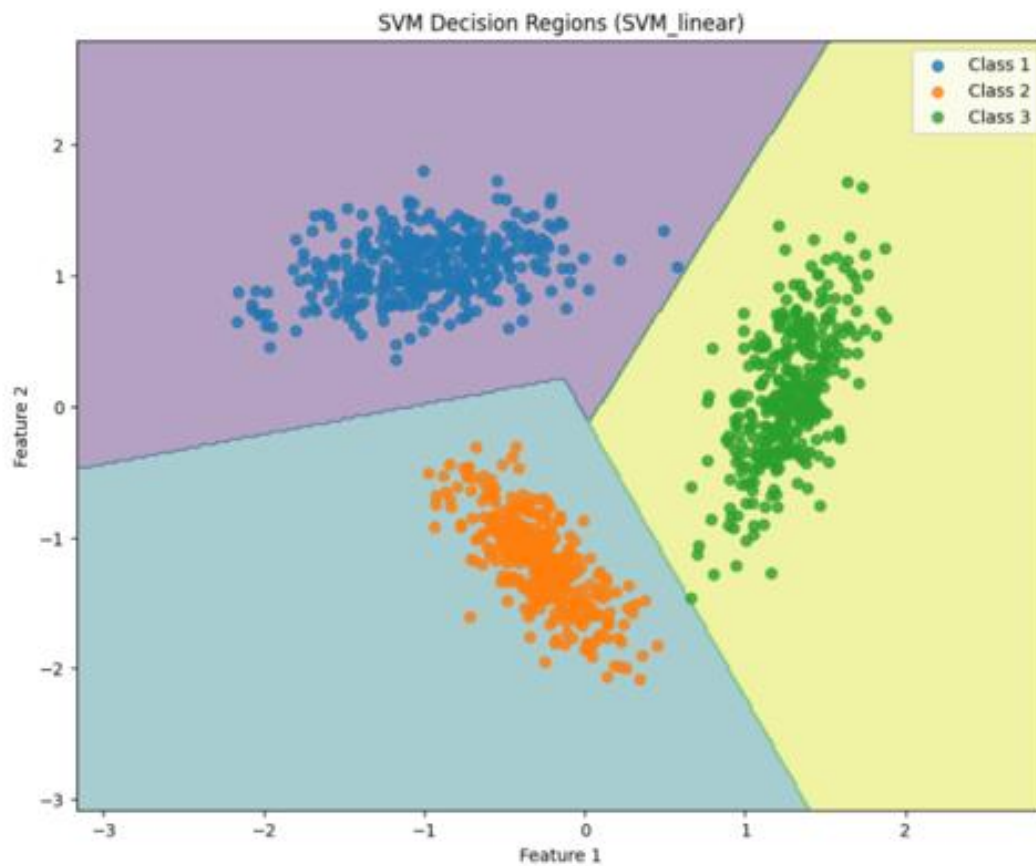


Figure 7 SVM Decision Regions

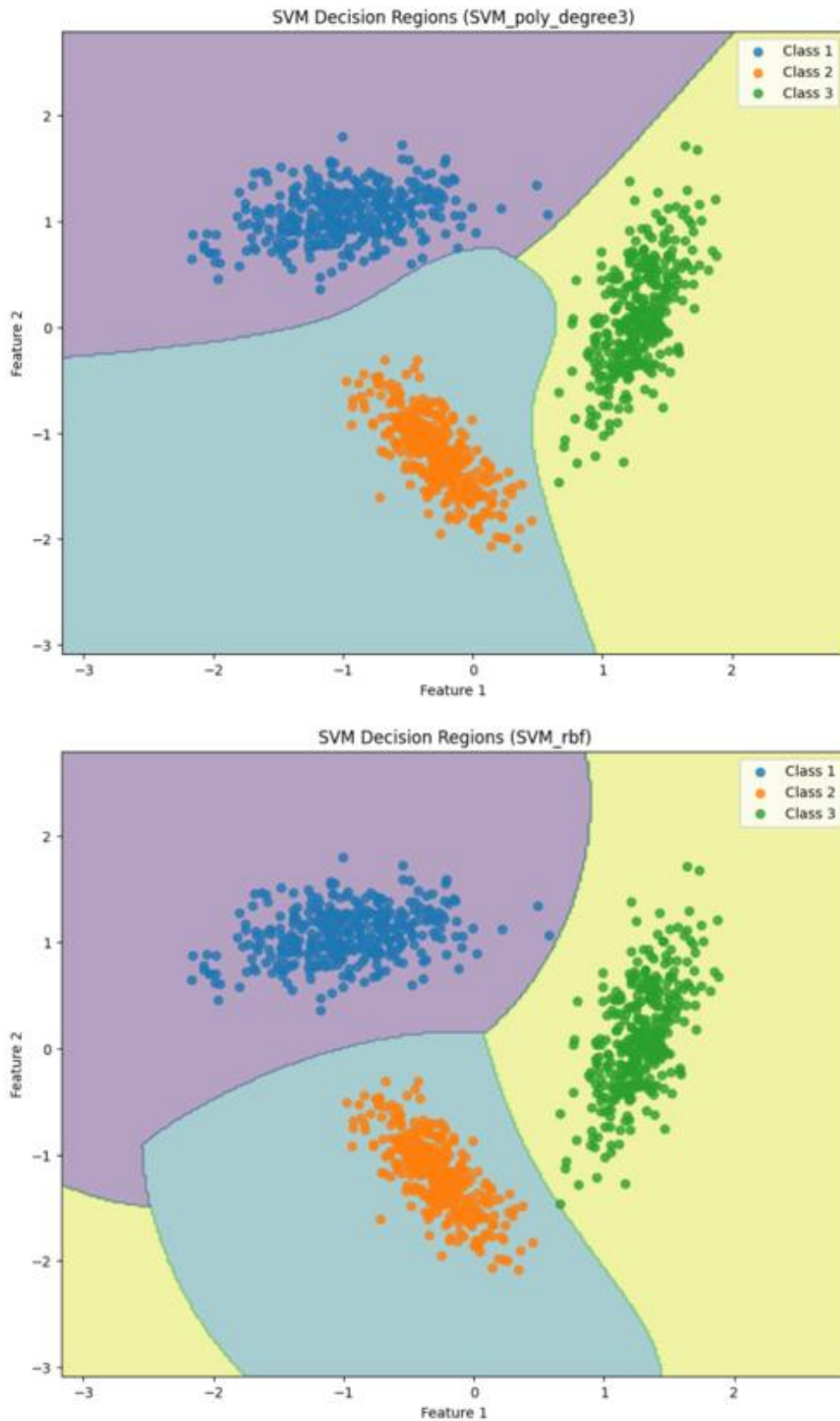


Figure 8 SVM Decision Regions

C. Plot Visualizations for Non-Linearly Separable Dataset

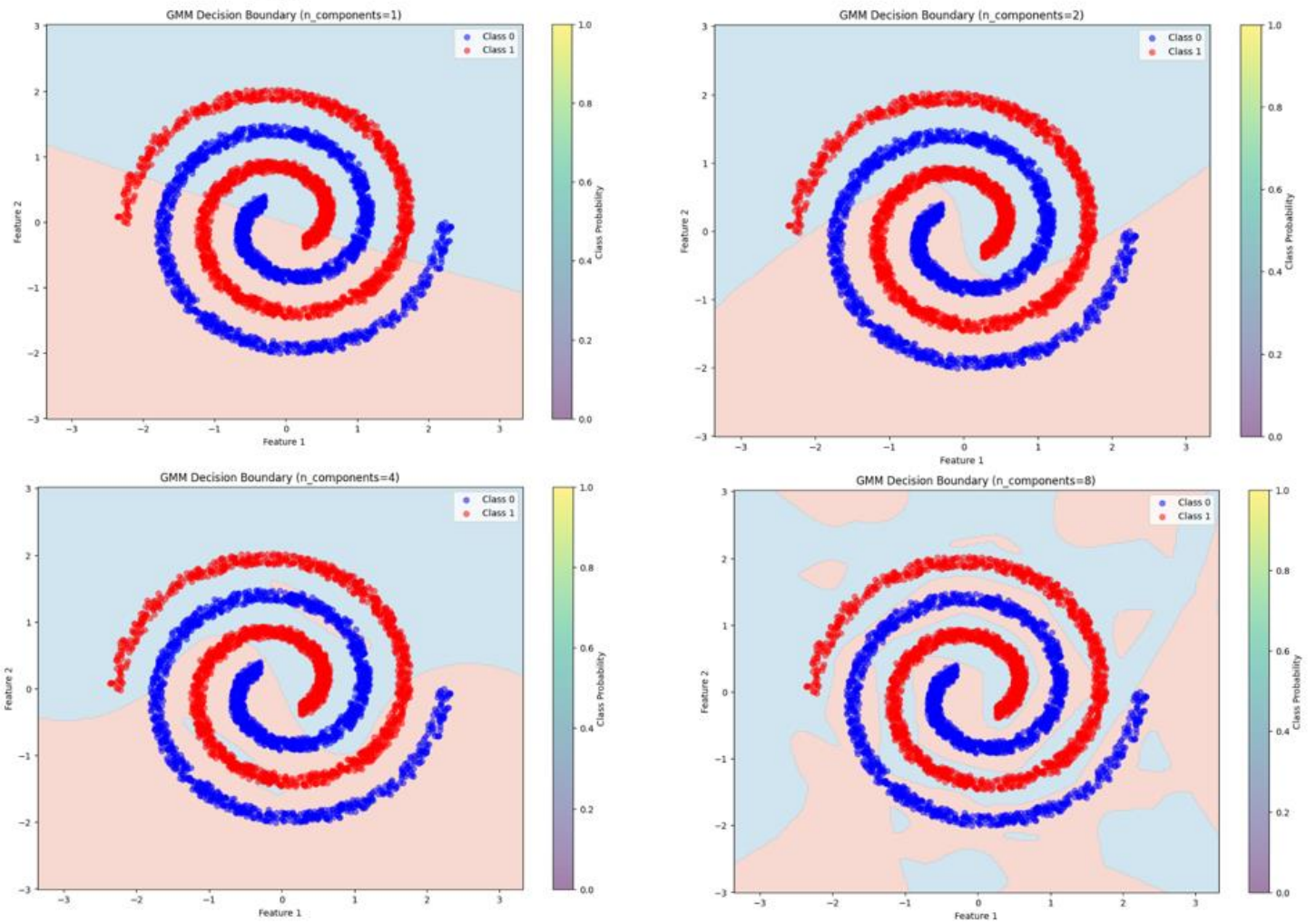


Figure 9 GMM Based Classifier Decision Regions for Components 1 to 8

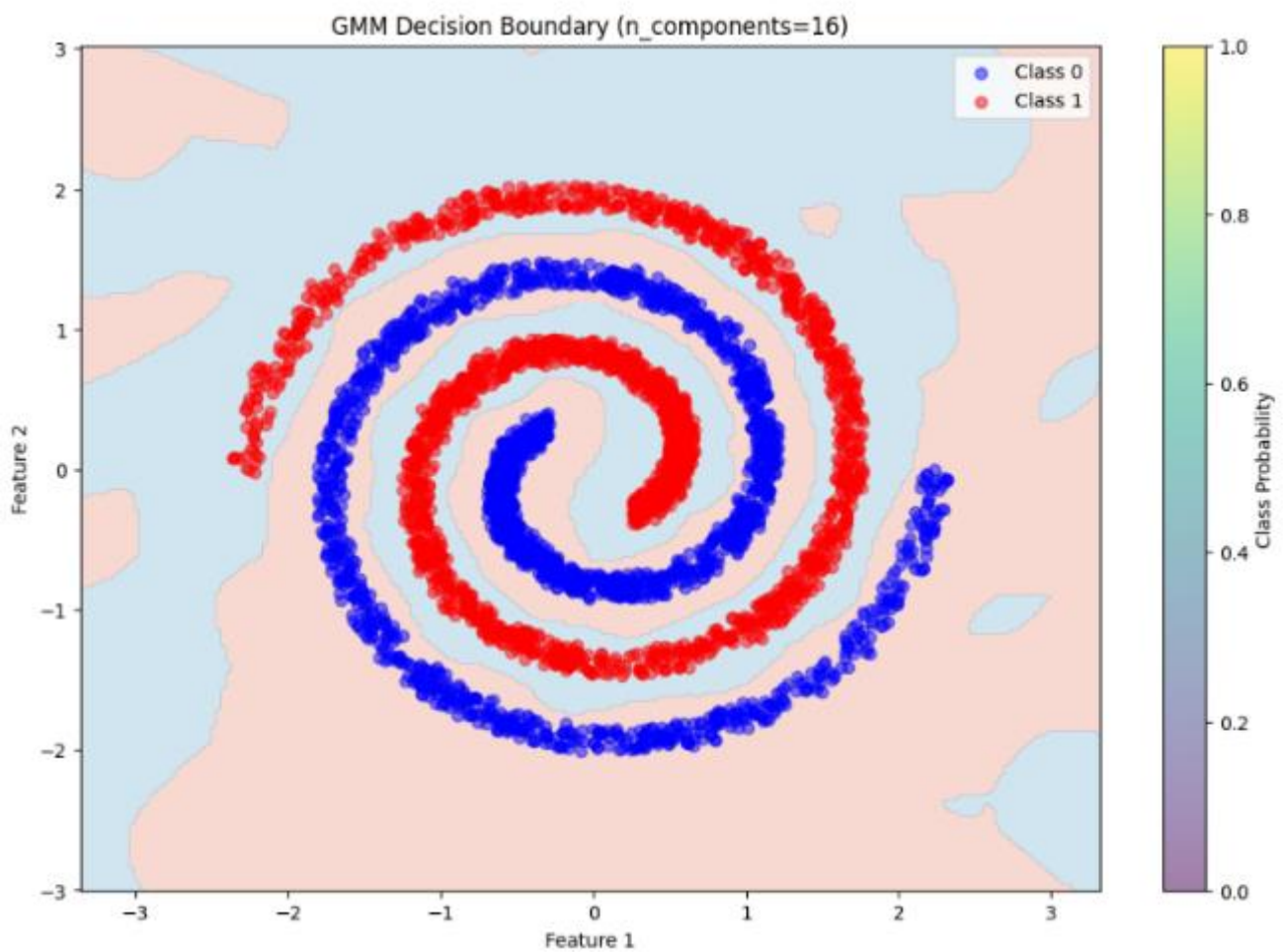


Figure 10 GMM Based Classifier Decision Regions for Components 16

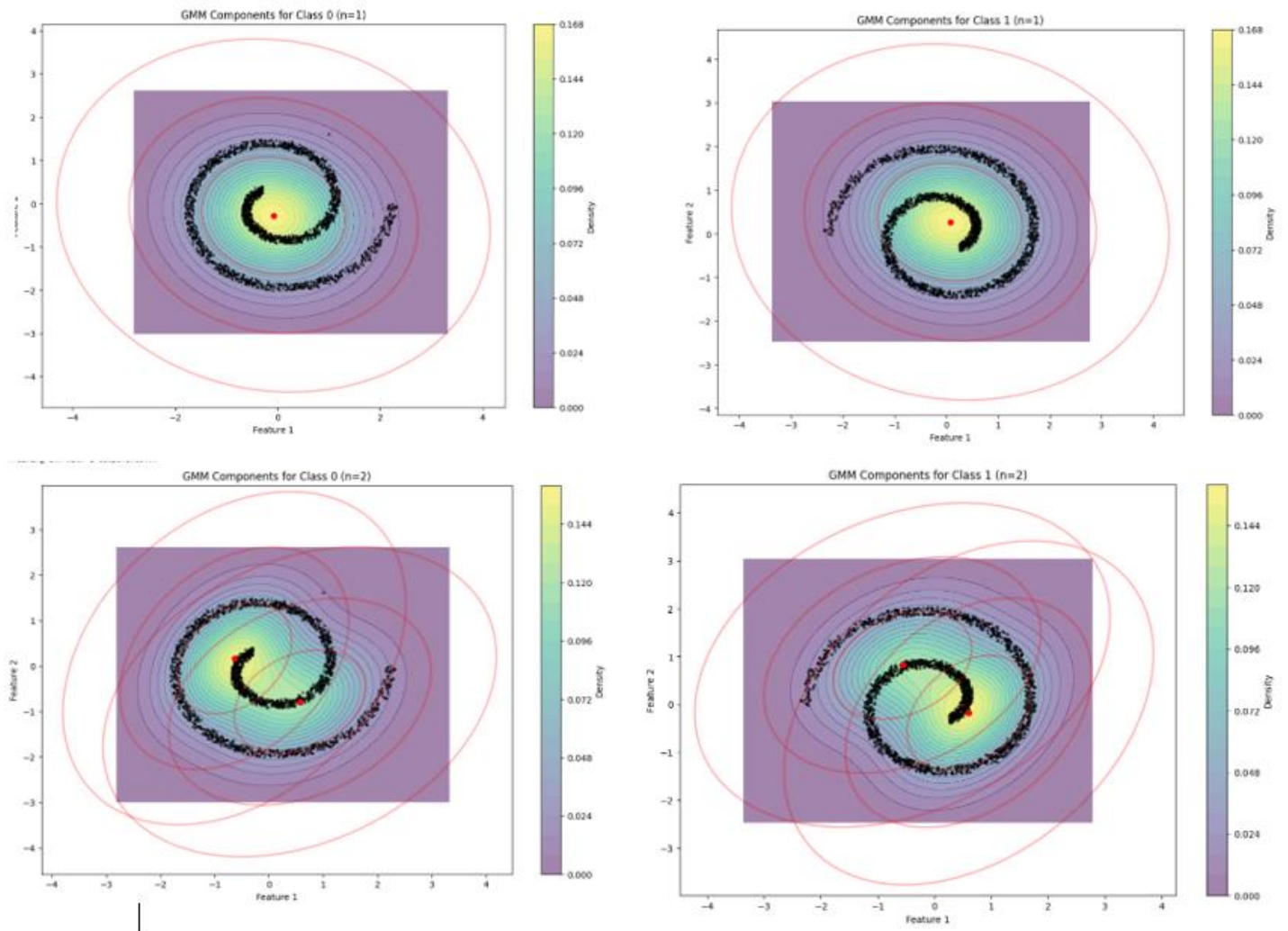


Figure 11 GMM Cluster centers for GMM Components 1 to 2

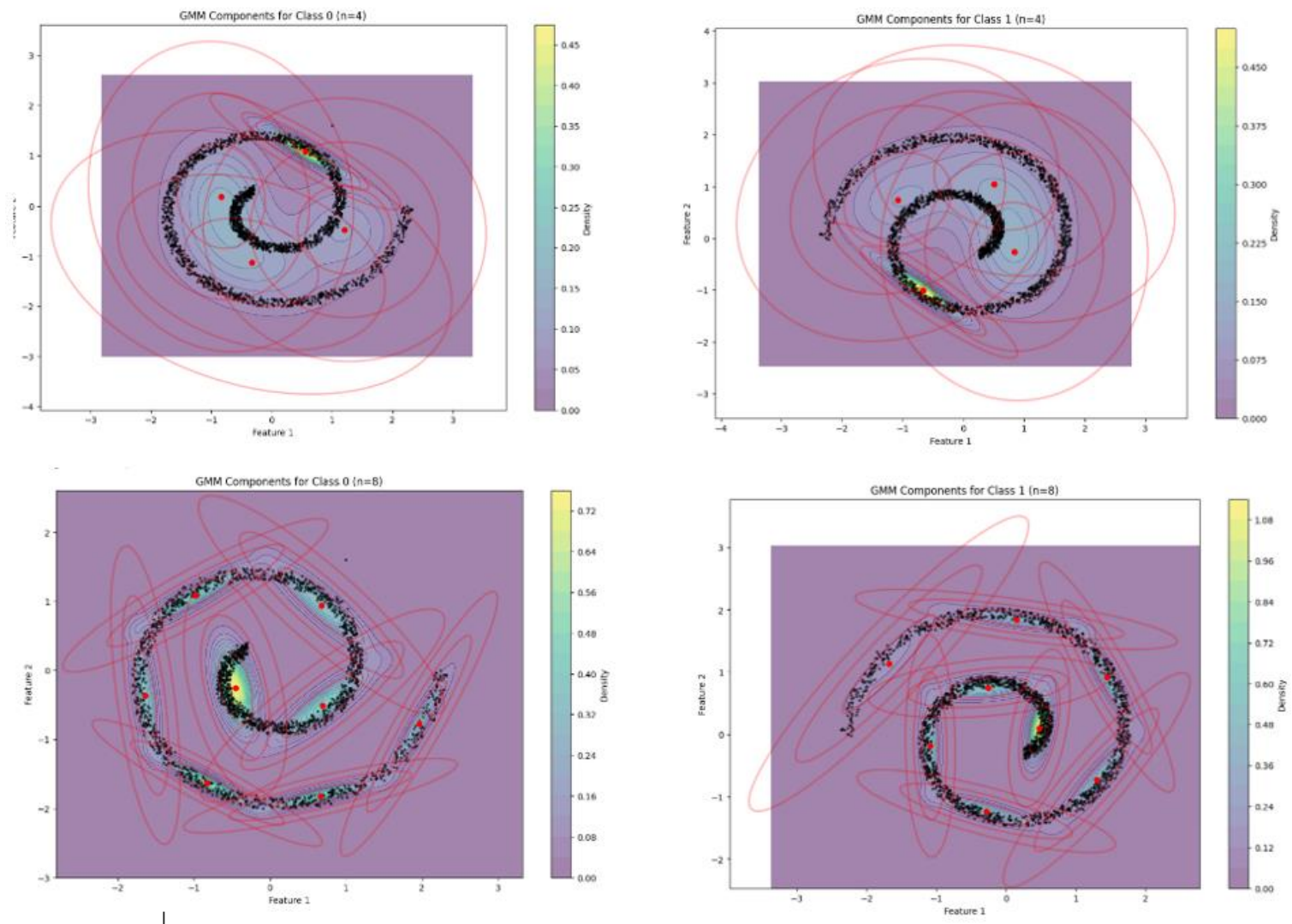


Figure 12 GMM Cluster centers for GMM Components 4 to 8

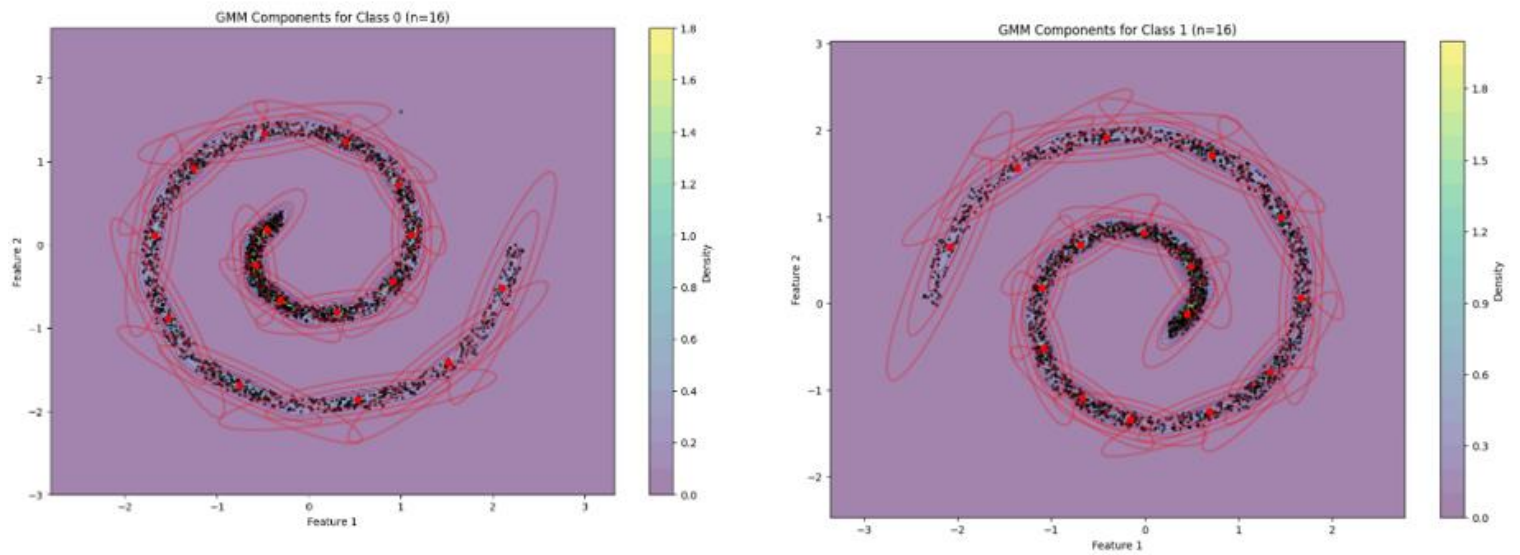


Figure 13 GMM Cluster centers for GMM Components 16

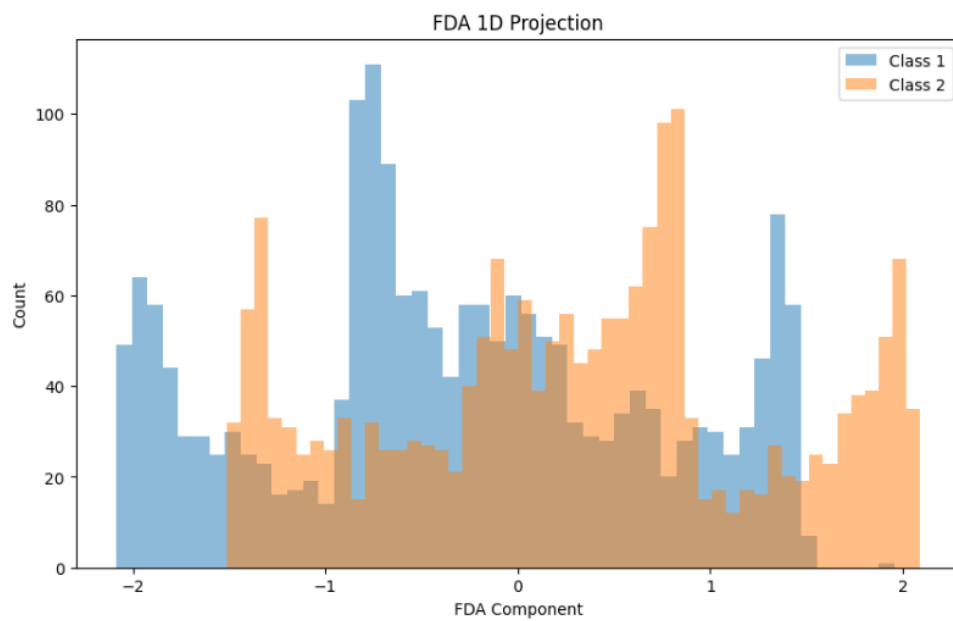


Figure 14 FDA 1D Projection

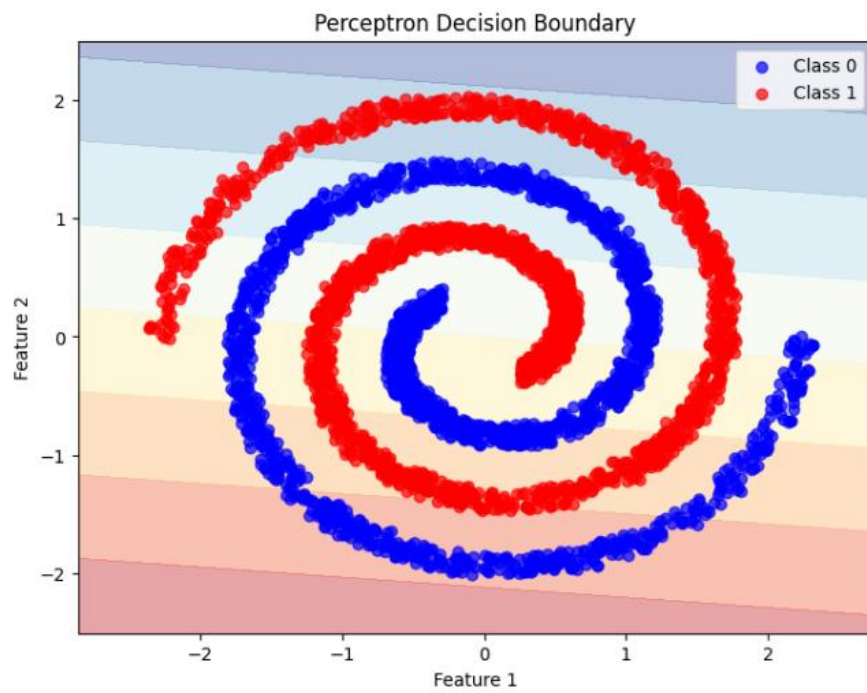


Figure 15 Perceptron Decision Boundary

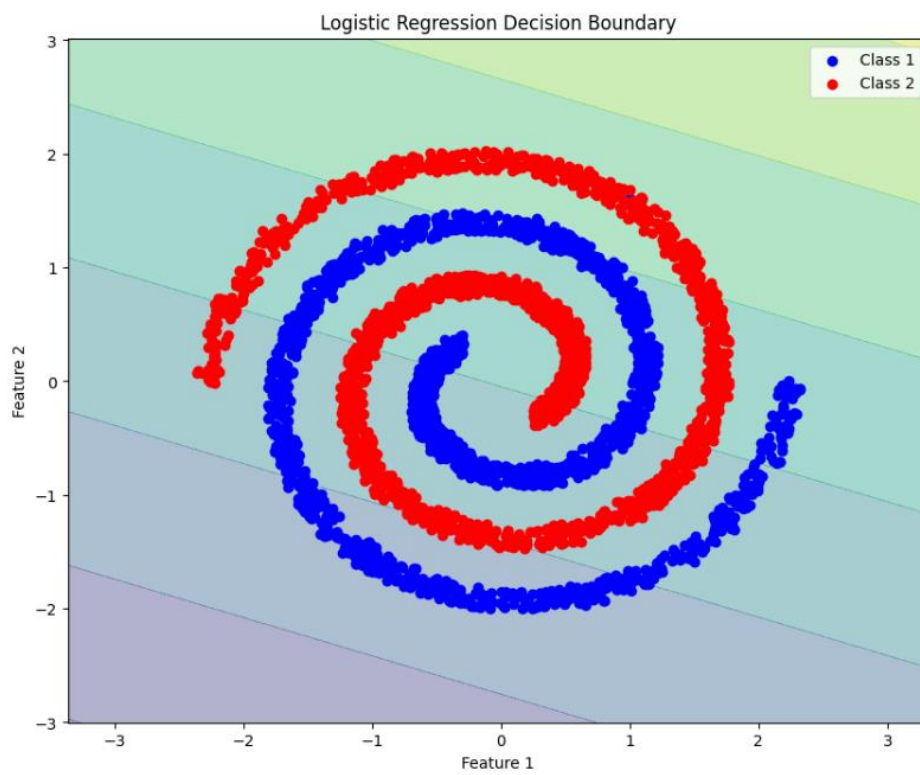


Figure 16 Logistic Regression Decision Boundary

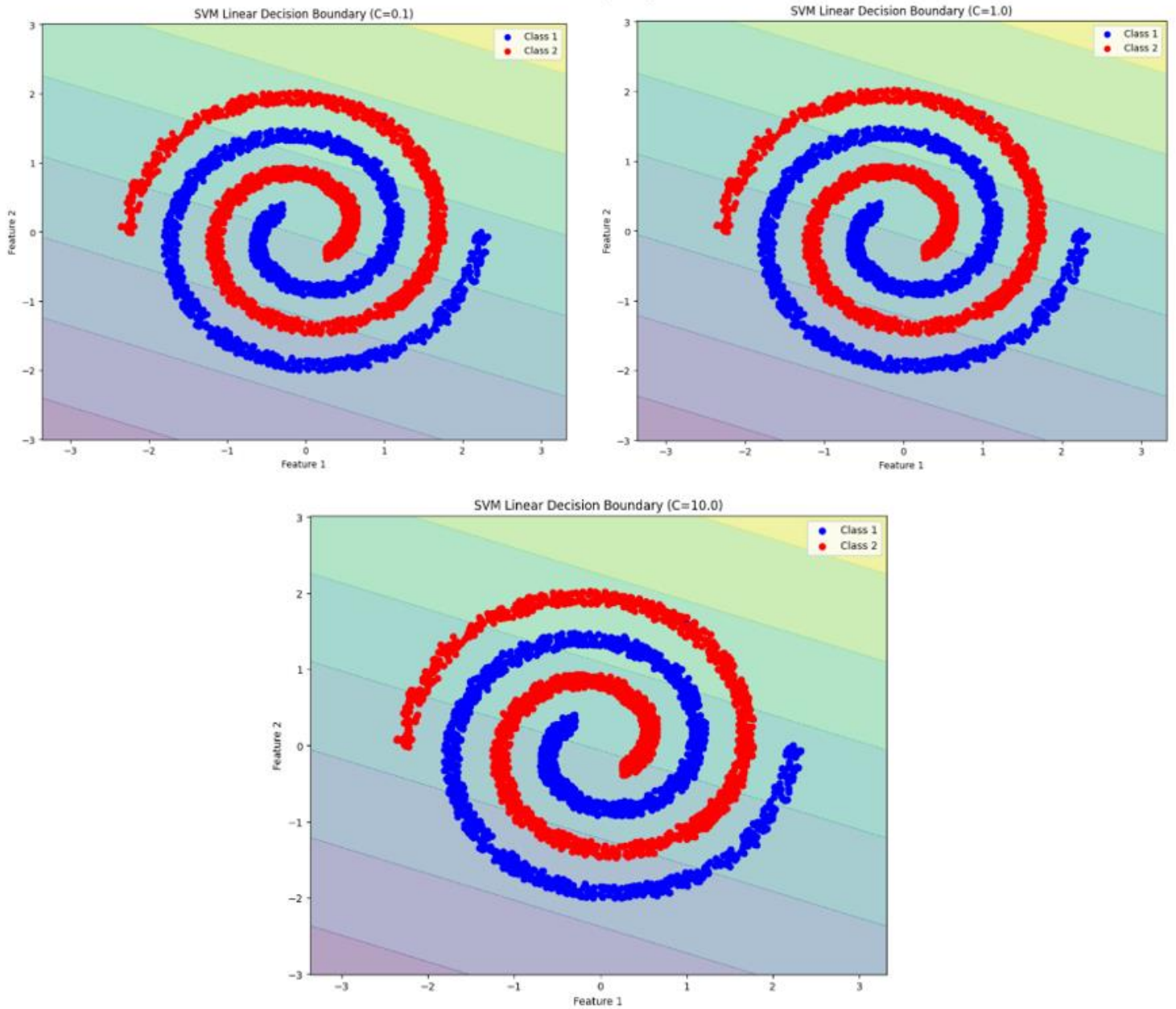


Figure 17 SVM Linear Decision Boundary

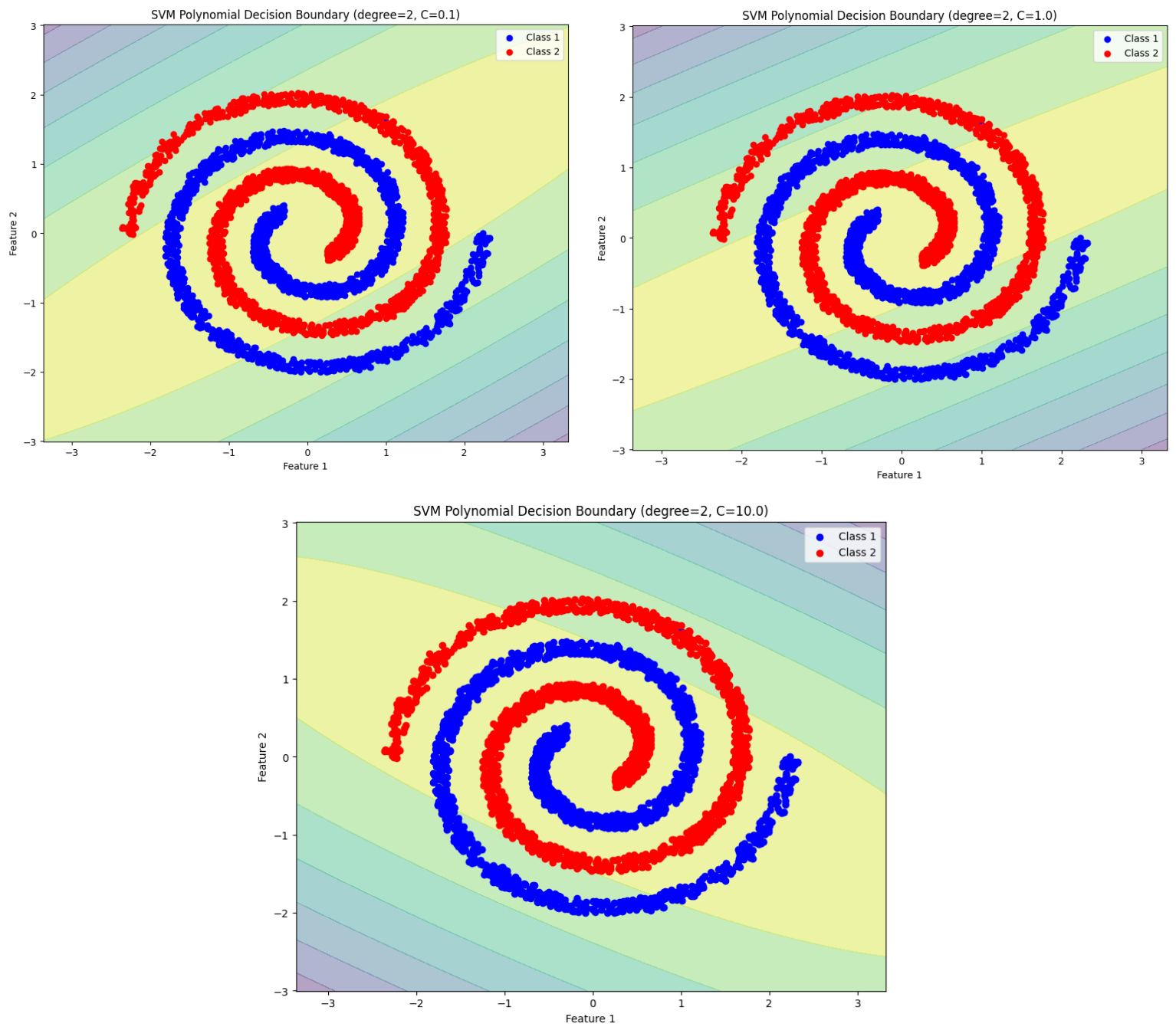


Figure 18 SVM Polynomial Decision Boundary for Degree = 2

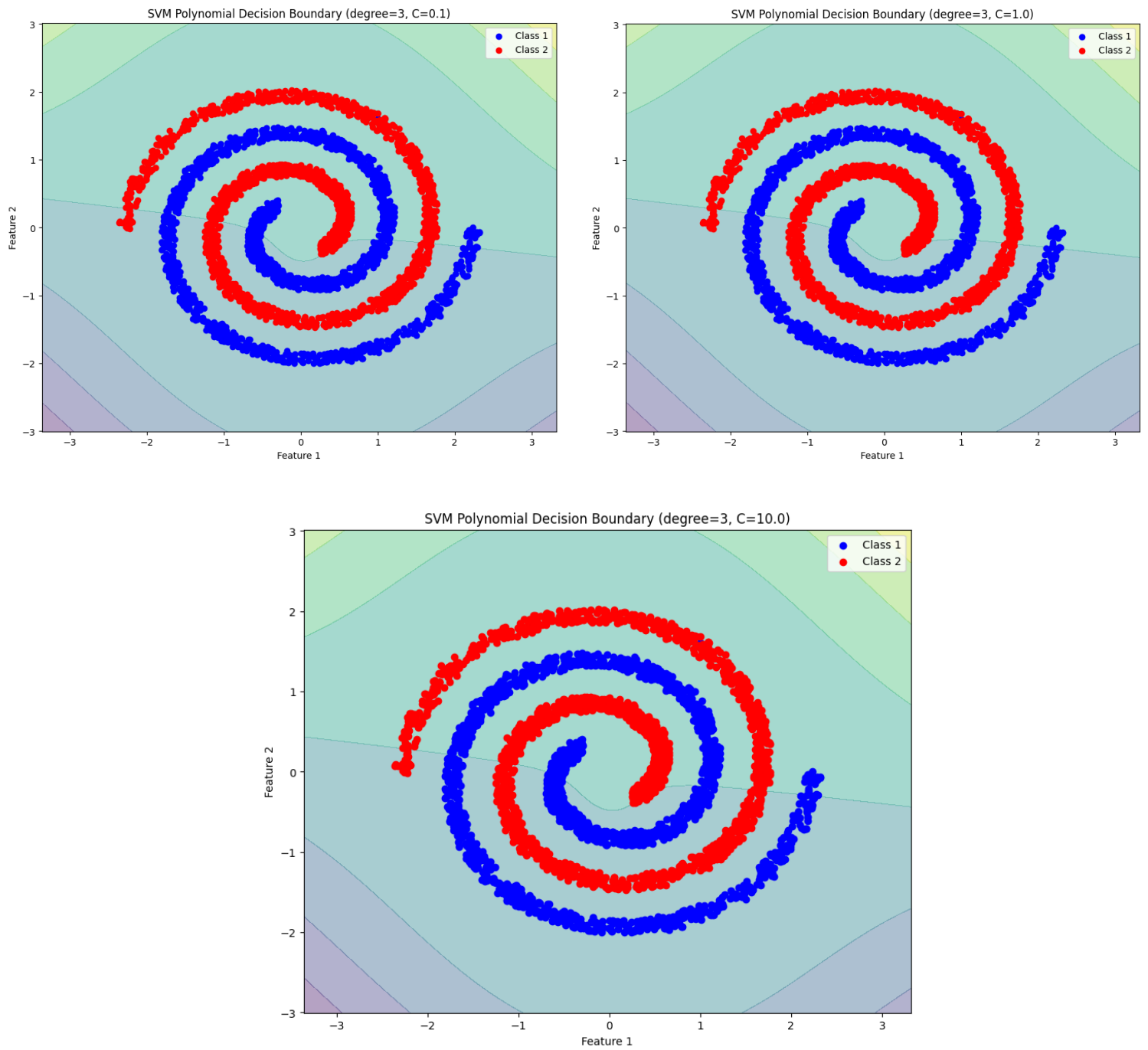


Figure 19 SVM Polynomial Decision Boundary for Degree = 3

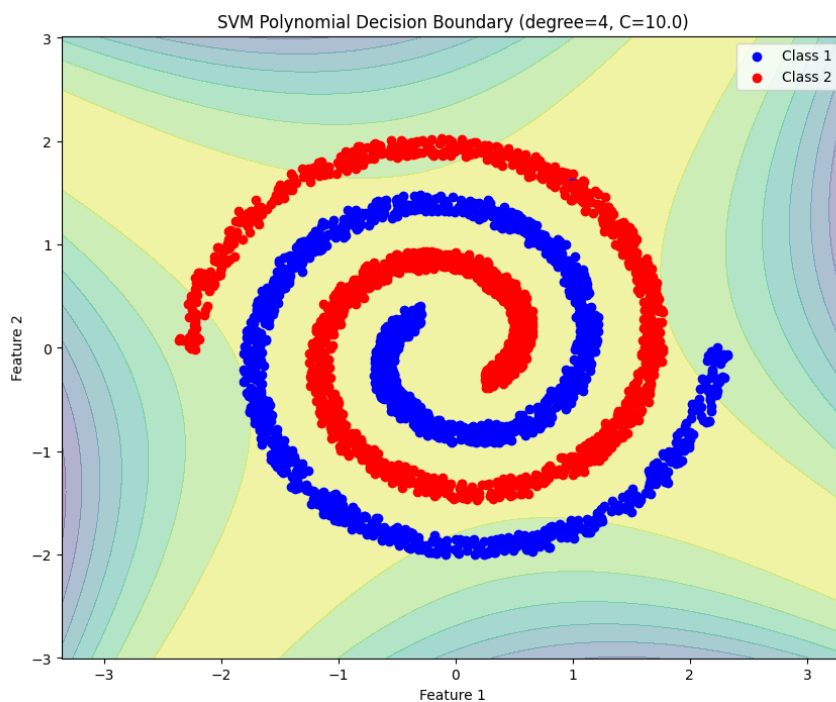
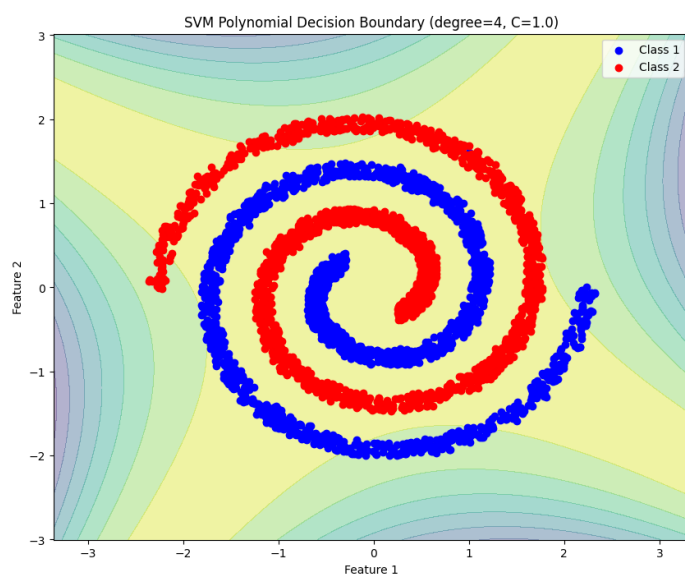
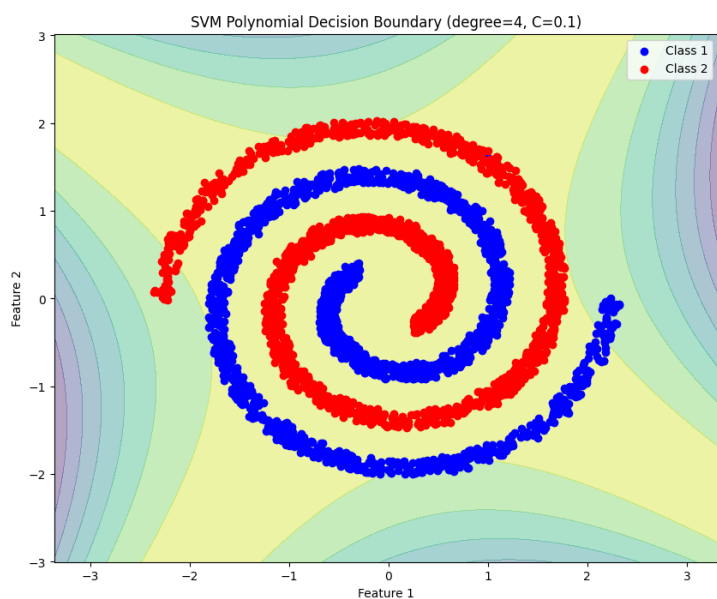


Figure 20 SVM Polynomial Decision Boundary for Degree = 4

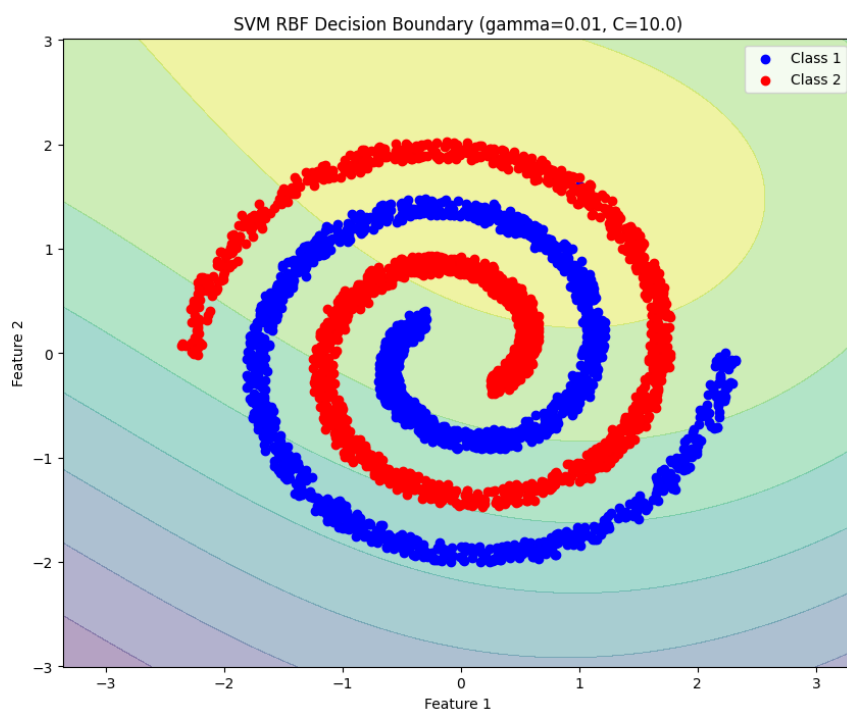
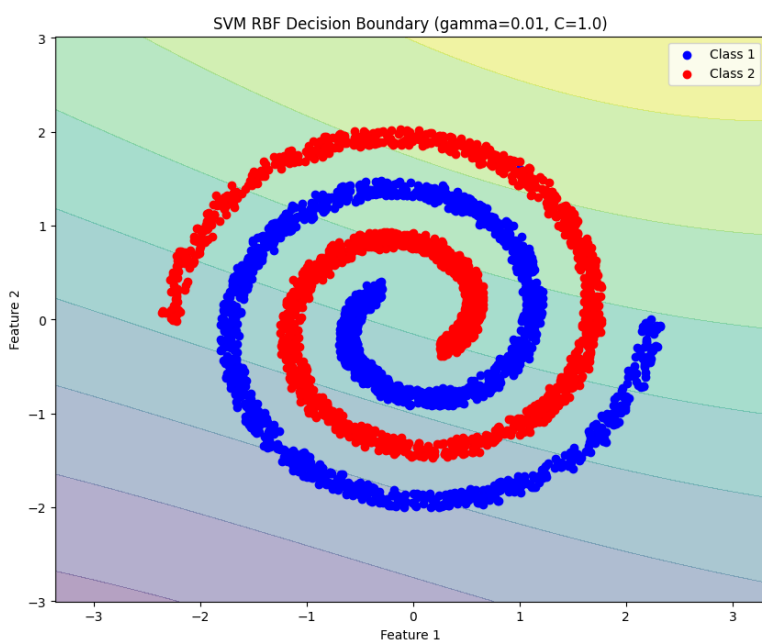
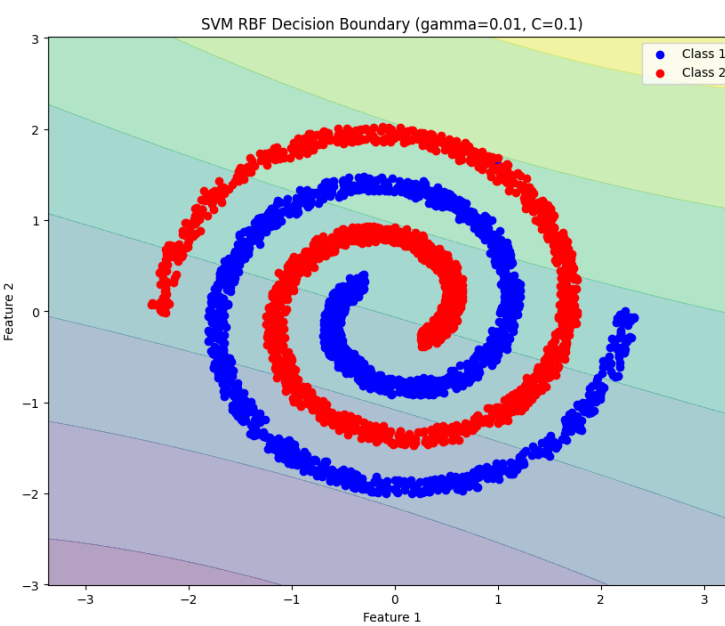


Figure 21 SVM RBF Decision Boundary (gamma =0.01)

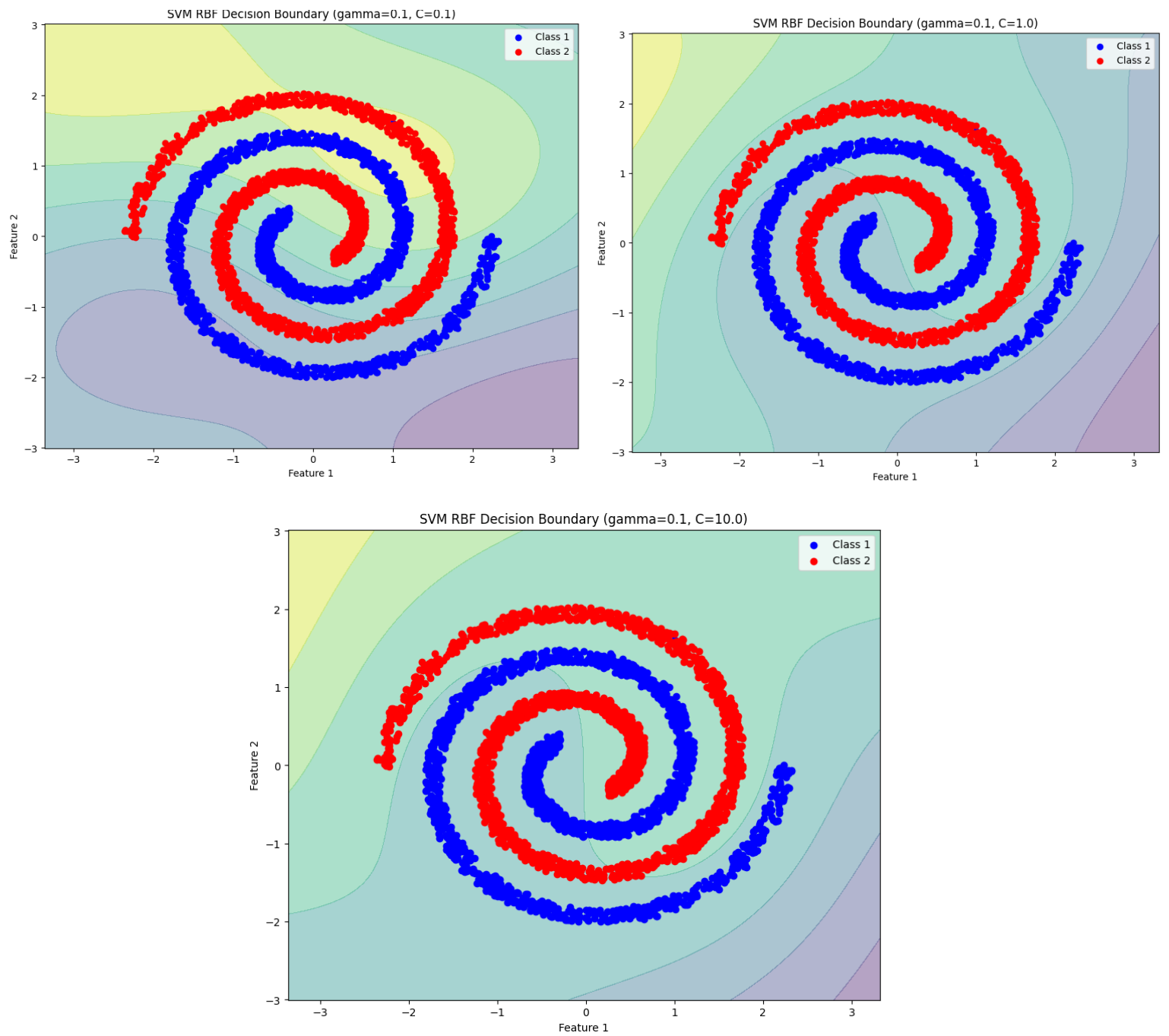


Figure 22 SVM RBF Decision Boundary (gamma =0.1)

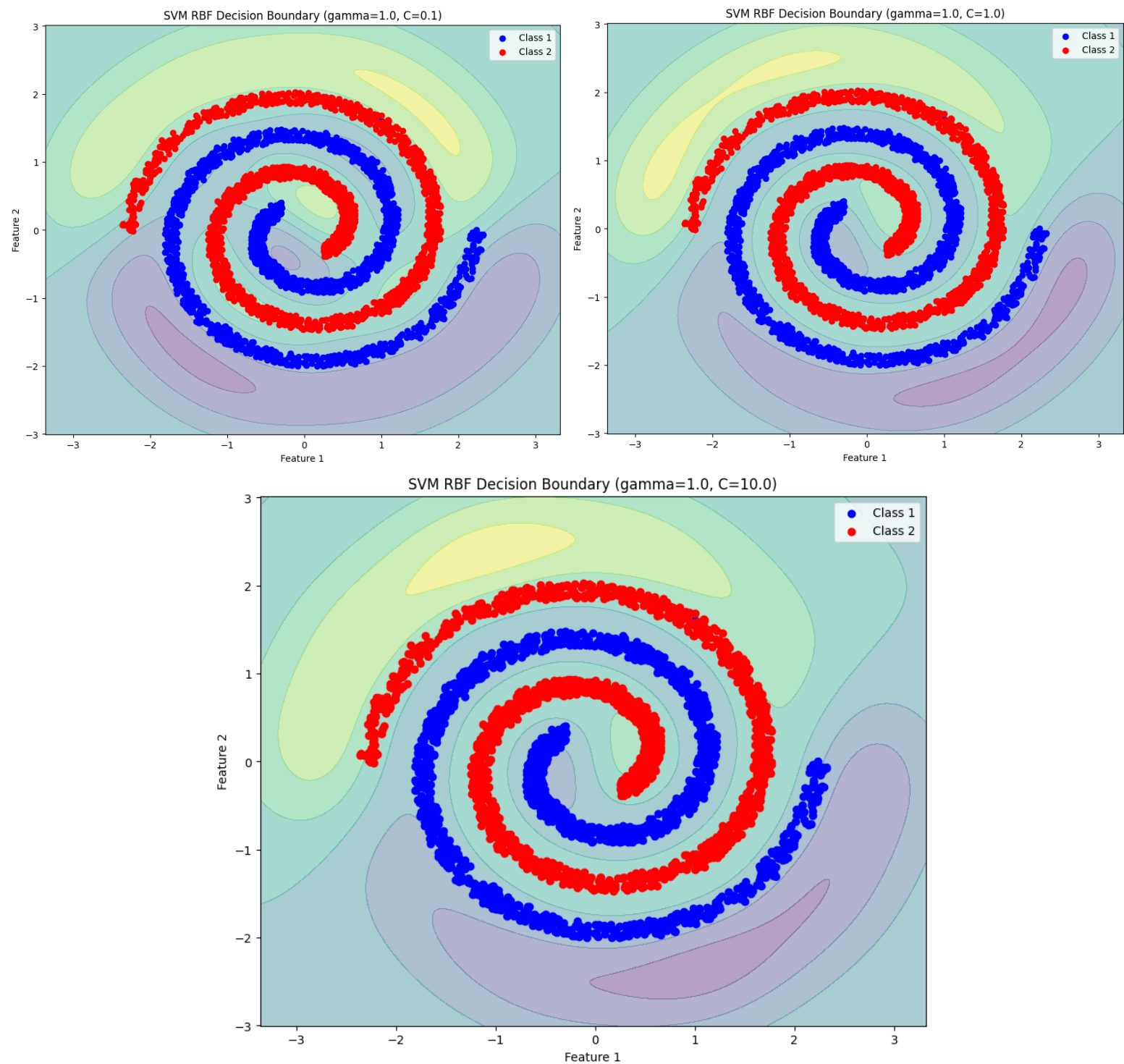


Figure 23 SVM RBF Decision Boundary ($\gamma=1$)

D. Results and Analysis

1. Classifier Performance Metrics for Linearly Separable dataset

Confusion Matrix and Analysis: GMM-based Bayes Classifiers GMM with 1,2,4,8 components:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 1:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 2:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 3:Result

Confusion Matrix and Analysis: KNN-based Bayes Classifiers KNN with K=1 ,3,5,7 :

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 4:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 5:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 6:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with Gaussian assumption:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 7:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 8:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 9:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with GMM:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 40:Confusion

Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 11:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 15:Result

Confusion Matrix and Analysis: Multiclass Perceptron Classifiers:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	1	149	0
Class3	0	0	150

Table 64:Confusion

Matrix

	Class1	Class2	Class3
Precision	0.9934	1.0000	1.0000
Recall	1.0000	0.9933	1.0000
F-Measure	0.9967	0.9967	1.0000

Table 15:Analysis

Accuracy	0.9978
Mean Precision	0.9978
Mean Recall	0.9978
Mean F-Measure	0.9978

Table 16:Result

Confusion Matrix and Analysis: Multiclass Logistic Regression:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	1	149

Table 77:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	0.9934	1.0000
Recall	1.0000	1.0000	0.9933
F-Measure	1.0000	0.9967	0.9967

Table 18:Analysis

Accuracy	0.9978
Mean Precision	0.9978
Mean Recall	0.9978
Mean F-Measure	0.9978

Table 19:Result

Confusion Matrix and Analysis: SVM Linear Classifiers:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 20:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 81:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 29:Result

Confusion Matrix and Analysis: SVM Poly_degree2 Classifiers:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	139	3	8
Class2	0	148	2
Class3	0	0	140

Table 24:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9329	0.9801	0.9333
Recall	0.9267	0.9867	0.9333
F-Measure	0.9298	0.9834	0.9333

Table 105:Analysis

Accuracy	0.9489
Mean Precision	0.9488
Mean Recall	0.9489
Mean F-Measure	0.9488

Table 26:Result

Confusion Matrix and Analysis: SVM Poly_degree3 Classifiers:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 27:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 118:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 29:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers:

	Actual		
Predicted	Class 1	Class 2	Class 3
Class1	150	0	0
Class2	0	150	0
Class3	0	0	150

Table 30:Confusion Matrix

	Class1	Class2	Class3
Precision	1.0000	1.0000	1.0000
Recall	1.0000	1.0000	1.0000
F-Measure	1.0000	1.0000	1.0000

Table 31:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 122:Result

2. Classifier Performance Metrics for Non Linearly Separable dataset

Confusion Matrix and Analysis: GMM-based Bayes Classifiers GMM with 1 component:

	Predicted	
Actual	Class1	Class2
Class1	314	201
Class2	162	302

Table 33:Confusion Matrix

	Class1	Class2
Precision	0.6597	0.6004
Recall	0.6097	0.6509
F-Measure	0.6337	0.6246

Table 34:Analysis

Accuracy	0.6292
Mean Precision	0.6300
Mean Recall	0.6303
Mean F-Measure	0.6291

Table 135:Result

Confusion Matrix and Analysis: GMM-based Bayes Classifiers GMM with 2 components:

	Predicted	
Actual	Class1	Class2
Class1	324	191
Class2	150	314

Table 36:Confusion Matrix

	Class1	Class2
Precision	0.6835	0.6218
Recall	0.6291	0.6767
F-Measure	0.6552	0.6481

Table 37:Analysis

Accuracy	0.6517
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 148:Result

Confusion Matrix and Analysis: GMM-based Bayes Classifiers GMM with 4 components:

	Predicted	
Actual	Class1	Class2
Class1	360	155
Class2	135	329

Table 39:Confusion Matrix

	Class1	Class2
Precision	0.7273	0.6798
Recall	0.6990	0.7091
F-Measure	0.7129	0.6941

Table 40:Analysis

Accuracy	0.7038
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 41:Result

Confusion Matrix and Analysis: GMM-based Bayes Classifiers GMM with 8 components:

	Predicted	
Actual	Class1	Class2
Class1	515	0
Class2	5	459

Table 42:Confusion Matrix

	Class1	Class2
Precision	0.9904	1.0000
Recall	1.0000	0.9892
F-Measure	0.9952	0.9946

Table 43:Analysis

Accuracy	0.9949
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 44:Result

Confusion Matrix and Analysis: GMM-based Bayes Classifiers GMM with 16 components:

	Predicted	
Actual	Class1	Class2
Class1	515	0
Class2	0	464

Table 45:Confusion Matrix

	Class1	Class2
Precision	1.0000	1.0000
Recall	1.0000	1.0000
F-Measure	1.0000	1.0000

Table 46:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 47:Result

Confusion Matrix and Analysis: KNN-based Bayes Classifiers KNN with K= 1:

	Predicted	
Actual	Class1	Class2
Class1	374	141
Class2	110	354

Table 48:Confusion Matrix

	Class1	Class2
Precision	0.7727	0.7152
Recall	0.7262	0.7629
F-Measure	0.7487	0.7383

Table 49:Analysis

Accuracy	0.7436
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 50:Result

Confusion Matrix and Analysis: KNN-based Bayes Classifiers KNN with K= 3:

	Predicted	
Actual	Class1	Class2
Class1	313	202
Class2	158	306

Table 515:Confusion Matrix

	Class1	Class2
Precision	0.6645	0.6024
Recall	0.6078	0.6595
F-Measure	0.6349	0.6296

Table 516:Analysis

Accuracy	0.6323
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 517:Result

Confusion Matrix and Analysis: KNN-based Bayes Classifiers KNN with K= 5:

	Predicted	
Actual	Class1	Class2
Class1	310	205
Class2	160	304

Table 54:Confusion Matrix

	Class1	Class2
Precision	0.6596	0.5972
Recall	0.6019	0.6552
F-Measure	0.6294	0.6249

Table 55:Analysis

Accuracy	0.6272
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 56:Result

Confusion Matrix and Analysis: KNN-based Bayes Classifiers KNN with K= 7:

	Predicted	
Actual	Class1	Class2
Class1	313	202
Class2	161	303

Table 57:Confusion Matrix

	Class1	Class2
Precision	0.6603	0.6000
Recall	0.6078	0.6530
F-Measure	0.6330	0.6254

Table 58:Analysis

Accuracy	0.6292
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 59:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with Gaussian assumption:

	Predicted	
Actual	Class1	Class2
Class1	302	213
Class2	156	308

Table 60:Confusion Matrix

	Class1	Class2
Precision	0.6594	0.5912
Recall	0.5864	0.6638
F-Measure	0.6208	0.6264

Table 61:Analysis

Accuracy	0.6231
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 62:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with GMM(components=1):

	Predicted	
Actual	Class1	Class2
Class1	314	201
Class2	162	302

Table 63:Confusion Matrix

	Class1	Class2
Precision	0.6597	0.6004
Recall	0.6097	0.6509
F-Measure	0.6337	0.6246

Table 64:Analysis

Accuracy	0.6292
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 65:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with GMM(components=2):

	Predicted	
Actual	Class1	Class2
Class1	293	222
Class2	135	339

Table 66:Confusion Matrix

	Class1	Class2
Precision	0.6846	0.5971
Recall	0.5689	0.7091
F-Measure	0.6214	0.6483

Table 67:Analysis

Accuracy	0.6393
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 68:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with GMM(components=4):

	Predicted	
Actual	Class1	Class2
Class1	350	165
Class2	123	341

Table 69:Confusion Matrix

	Class1	Class2
Precision	0.7400	0.6739
Recall	0.6796	0.7349
F-Measure	0.7085	0.7031

Table 70:Analysis

Accuracy	0.7058
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 71:Result

Confusion Matrix and Analysis: FDA-based Bayes Classifiers FDA with GMM(components=8):

	Predicted	
Actual	Class1	Class2
Class1	355	160
Class2	133	331

Table 72:Confusion Matrix

	Class1	Class2
Precision	0.7275	0.6741
Recall	0.6893	0.7134
F-Measure	0.7079	0.6932

Table 73:Analysis

Accuracy	0.7007
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 74:Result

Confusion Matrix and Analysis: Multiclass Perceptron Classifiers:

	Predicted	
Actual	Class1	Class2
Class1	79	436
Class2	3	461

Table 75:Confusion Matrix

	Class1	Class2
Precision	0.9634	0.5139
Recall	0.1534	0.9935
F-Measure	0.2647	0.6774

Table 76:Analysis

Accuracy	0.5516
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 77:Result

Confusion Matrix and Analysis: Multiclass Logistic Regression:

	Predicted	
Actual	Class1	Class2
Class1	302	213
Class2	156	308

Table 78:Confusion Matrix

	Class1	Class2
Precision	0.6594	0.5912
Recall	0.5864	0.6638
F-Measure	0.6208	0.6254

Table 79:Analysis

Accuracy	0.6231
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 80:Result

Confusion Matrix and Analysis: SVM Linear Classifiers(c=0.1):

	Predicted	
Actual	Class1	Class2
Class1	300	215
Class2	154	310

Table 81:Confusion Matrix

	Class1	Class2
Precision	0.6608	0.5905
Recall	0.5825	0.6681
F-Measure	0.6192	0.6269

Table 818:Analysis

Accuracy	0.6231
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 819:Result

Confusion Matrix and Analysis: SVM Linear Classifiers(c=1):

	Predicted	
Actual	Class1	Class2
Class1	301	214
Class2	154	310

Table 84:Confusion Matrix

	Class1	Class2
Precision	0.6615	0.5916
Recall	0.5845	0.6681
F-Measure	0.6206	0.6275

Table 85:Analysis

Accuracy	0.6241
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 86:Result

Confusion Matrix and Analysis: SVM Linear Classifiers(c=10):

	Predicted	
Actual	Class1	Class2
Class1	301	214
Class2	154	310

Table 87:Confusion Matrix

	Class1	Class2
Precision	0.6615	0.5916
Recall	0.5845	0.6681
F-Measure	0.6206	0.6275

Table 88:Analysis

Accuracy	0.6241
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 89:Result

Confusion Matrix and Analysis: SVM Poly_degree2 Classifiers(c= 0.1 , 1 , 10):

	Predicted	
Actual	Class1	Class2
Class1	0	515
Class2	0	464

Table 90:Confusion Matrix

	Class1	Class2
Precision	0.0000	0.4740
Recall	0.0000	1.0000
F-Measure	0.0000	0.6231

Table 91:Analysis

Accuracy	0.4740
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 92:Result

Confusion Matrix and Analysis: SVM Poly_degree3 Classifiers (c=0.1) :

	Predicted	
Actual	Class1	Class2
Class1	305	210
Class2	127	337

Table 93:Confusion Matrix

	Class1	Class2
Precision	0.7060	0.6161
Recall	0.5922	0.7263
F-Measure	0.6441	0.6667

Table 94:Analysis

Accuracy	0.6558
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 95:Result

Confusion Matrix and Analysis: SVM Poly_degree3 Classifiers (c=1 , 10) :

	Predicted	
Actual	Class1	Class2
Class1	305	210
Class2	128	336

Table 96:Confusion Matrix

	Class1	Class2
Precision	0.7044	0.6154
Recall	0.5922	0.7241
F-Measure	0.6435	0.6653

Table 98:Analysis

Accuracy	0.6547
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 99:Result

Confusion Matrix and Analysis: SVM Poly_degree4 Classifiers (c=0.1) :

	Predicted	
Actual	Class1	Class2
Class1	20	495
Class2	39	425

Table 100:Confusion Matrix

	Class1	Class2
Precision	0.3390	0.4620
Recall	0.0388	0.9159
F-Measure	0.0697	0.6142

Table 101:Analysis

Accuracy	0.4545
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 102:Result

Confusion Matrix and Analysis: SVM Poly_degree4 Classifiers (c=1) :

	Predicted	
Actual	Class1	Class2
Class1	38	477
Class2	56	408

Table 103:Confusion Matrix

	Class1	Class2
Precision	0.4043	0.4610
Recall	0.0738	0.8793
F-Measure	0.1248	0.6049

Table 104:Analysis

Accuracy	0.4556
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 105:Result

Confusion Matrix and Analysis: SVM Poly_degree4 Classifiers (c=10) :

	Predicted	
Actual	Class1	Class2
Class1	39	476
Class2	56	408

Table 106:Confusion Matrix

	Class1	Class2
Precision	0.4105	0.4615
Recall	0.0757	0.8793
F-Measure	0.1279	0.6053

Table 107:Analysis

Accuracy	0.4566
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 108:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=0.01, C=0.1):

	Predicted	
Actual	Class1	Class2
Class1	300	215
Class2	155	309

Table 109:Confusion Matrix

	Class1	Class2
Precision	0.6593	0.5897
Recall	0.5825	0.6659
F-Measure	0.6186	0.6255

Table 110:Analysis

Accuracy	0.6221
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 112:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=0.01, C=1):

	Predicted	
Actual	Class1	Class2
Class1	293	222
Class2	138	326

Table 113:Confusion Matrix

	Class1	Class2
Precision	0.6798	0.5949
Recall	0.5689	0.7026
F-Measure	0.6195	0.6443

Table 114:Analysis

Accuracy	0.6323
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 115:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=0.01, C=10):

	Predicted	
Actual	Class1	Class2
Class1	222	293
Class2	114	350

Table 116:Confusion Matrix

	Class1	Class2
Precision	0.6607	0.5443
Recall	0.4311	0.7543
F-Measure	0.5217	0.6323

Table 117:Analysis

Accuracy	0.5843
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 118:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=0.1, C=0.1):

	Predicted	
Actual	Class1	Class2
Class1	283	232
Class2	125	339

Table 119:Confusion Matrix

	Class1	Class2
Precision	0.6636	0.5937
Recall	0.5495	0.7306
F-Measure	0.6132	0.6551

Table 120:Analysis

Accuracy	0.6353
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 121:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=0.1, C=1):

	Predicted	
Actual	Class1	Class2
Class1	341	274
Class2	135	329

Table 122:Confusion Matrix

	Class1	Class2
Precision	0.7164	0.6541
Recall	0.6621	0.7091
F-Measure	0.6882	0.6805

Table 123:Analysis

Accuracy	0.6844
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 124:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=0.1, C=10):

	Predicted	
Actual	Class1	Class2
Class1	358	157
Class2	128	336

Table 125:Confusion Matrix

	Class1	Class2
Precision	0.7366	0.6815
Recall	0.6951	0.7241
F-Measure	0.7153	0.7022

Table 126:Analysis

Accuracy	0.7089
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 127:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=1, C=0.1):

	Predicted	
Actual	Class1	Class2
Class1	453	62
Class2	39	425

Table 128:Confusion Matrix

	Class1	Class2
Precision	0.9207	0.8727
Recall	0.8796	0.9159
F-Measure	0.8997	0.8938

Table 129:Analysis

Accuracy	0.8968
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 130:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=1, C=1):

	Predicted	
Actual	Class1	Class2
Class1	512	3
Class2	0	464

Table 131:Confusion Matrix

	Class1	Class2
Precision	1.0000	0.9936
Recall	0.9942	1.0000
F-Measure	0.9971	0.9968

Table 132:Analysis

Accuracy	0.9969
Mean Precision	
Mean Recall	
Mean F-Measure	

Table 133:Result

Confusion Matrix and Analysis: SVM_rbf Classifiers(gamma=1, C=10):

	Predicted	
Actual	Class1	Class2
Class1	515	0
Class2	0	464

Table 134:Confusion Matrix

	Class1	Class2
Precision	1.0000	1.0000
Recall	1.0000	1.0000
F-Measure	1.0000	1.0000

Table 135:Analysis

Accuracy	1.0000
Mean Precision	1.0000
Mean Recall	1.0000
Mean F-Measure	1.0000

Table 136:Result

- Classification accuracy.
- Per-class and mean precision, recall, and F-measure.
- Confusion matrices for test datasets.

2. KNN-Based Bayes Classifier Analysis

- Observations:
 - Increasing K resulted in smoother decision boundaries but reduced flexibility.
 - Optimal performance observed at K=3.

3. FDA Analysis

- **Unimodal Gaussian:**

- Performed well on linearly separable data but struggled with nonlinear patterns.
- **GMM:**
 - Improved flexibility, capturing non-linear class structures effectively.

4. Perceptron Analysis

- Convergence was achieved faster for linearly separable data.
- Performance declined significantly on nonlinearly separable data.

5. Logistic Regression Analysis

- Showed competitive performance with smooth decision boundaries.
- Regularization helped control overfitting on nonlinear data.

6. SVM Analysis

- **Kernel Comparisons:**
 - Linear kernel excelled on linearly separable data.
 - Gaussian kernel outperformed on non-linear data, especially with optimized bandwidth.
 - Polynomial kernel provided a balance for moderately non-linear data.
- Optimal parameters were determined through grid search.

C. Comparative Analysis

1. Decision Region Plots

- KNN, perceptron, logistic regression, and SVM boundaries compared visually.
- Highlighted differences in decision boundaries for various methods.

2. Special Comparisons

- **Perceptron vs Linear SVM:**
 - SVM showed superior generalization on test data.
- **FDA Dimensionality Reduction:**
 - Effectively reduced dimensions for visualization and improved classifier performance.

III. Dataset 2: 3-Class Scene Image Dataset

A. Methodology

1. Bayes Classifier Using GMM

- GMMs with 1, 2, 4, and 8 mixtures were applied to the 32-dimensional BoVW dataset reduced via PCA.
- PCA reduced the dataset dimensions to 2, 5, 10, 15, and 20, retaining significant variance.
- The classification was conducted on the reduced dataset using GMM, and the effects of varying mixtures and PCA dimensions were analyzed.

2. Bayes Classifier Using KNN

- Experimented with $K=1, 3, 5$ for density estimation.
- Performance trends were analyzed.

3. FDA Implementation

- Dimensionality reduction via FDA.
- Comparison of unimodal Gaussian and GMM-based classifiers.

4. Perceptron Implementation

- Adapted perceptron for multi-class classification.

5. Logistic Regression Implementation

- Extended logistic regression for multi-class classification.

6. SVM Implementation

- Implemented SVM with one-vs-one and one-vs-all strategies for multi-class classification.
- Kernel and parameter tuning.

B. Results and Analysis

1. Solution Approach:

➤ Principle Components Analysis:

Principal component analysis (PCA) is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables (entities, each of which takes on various numerical values) into a set of values of linearly uncorrelated variables called principal components.

Procedure:

1. Take all the training examples of all the classes.
2. Compute the co-variance matrix.
3. Perform eigen analysis to obtain d eigenvalues and corresponding eigenvectors where d is the dimension of one feature vector.
4. Choose L eigen vectors corresponding to the L leading eigen values where L is the number of dimensions in which we our data is reduced to from d dimensions.
5. Project each of the data points (training as well as test) onto these L eigenvectors to obtain the L principal components.

After reducing the dimension of the 32-dimensional BoVW representation of our images, GMM based classifications was carried out on the reduced dimensional dataset. We have carried out the GMM based classification for 1; 2; 4; 8 mixture components.

a) Principle Component Analysis

➤ Variation Of Eigen Values

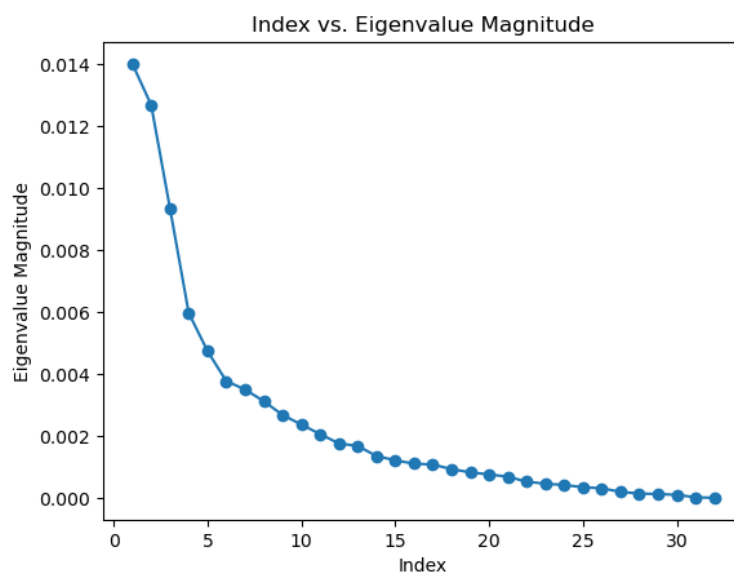


Figure 24 Variation of Eigen values

We observe that around 5 - 20 eigenvalues represent almost 95 percent of variation in our data, while other eigenvalues don't contribute much. We take the most significant L eigen vectors corresponding to the most significant L eigen values. The values of L considered are 2, 5, 10, 15, 21.

➤ **3.2 Confusion Matrix, Precision, Recall and F-measure**

➤ Where **Class1** -botanical_garden, **Class2** -bus_interior and **Class3** -elevator_shaft

1. Confusion Matrix and Analysis: When L = 1, M = 1

	Actual		
Predicted	Class1	Class2	Class3
Class1	34	16	0
Class2	12	32	6
Class3	13	36	1

Table 20:Confusion Matrix

	Class1	Class2	Class3
Precision	0.5762	0.3809	0.1428
Recall	0.68	0.64	0.02
F-Measure	0.6238	0.4776	0.0350

Table 21:Analysis

Accuracy	44.66%
Mean Precision	0.3666
Mean Recall	0.4466
Mean F-Measure	0.3788

Table 22:Result

➤ **Misclassified Images:**

True: botanical_garden, Predicted: bus_interior



Figure 26 Misclassified image as bus interior

True: botanical_garden, Predicted: bus_interior



Figure 25 Misclassified Image as bus interior

2. Confusion Matrix and Analysis: When L = 1, M = 2

	Actual		
Predicted	Class1	Class2	Class3
Class1	30	11	9
Class2	8	27	15
Class3	11	27	12

Table 4:Confusion Matrix

	Class1	Class2	Class3
Precision	0.6122	0.4153	0.3333
Recall	0.60	0.54	0.24
F-Measure	0.6060	0.4695	0.2790

Table 5:Analysis

Accuracy	46.00%
Mean Precision	0.4536
Mean Recall	0.46
Mean F-Measure	0.4515

Table 6:Result

➤ **Misclassified Images:**

True: botanical_garden, Predicted: bus_interior



Figure 28 Misclassified image as bus interior

True: botanical_garden, Predicted: elevator_shaft



Figure 27 Misclassified image as elevator shaft

3. Confusion Matrix and Analysis: When L = 1, M = 4

	Actual		
Predicted	Class1	Class2	Class3
Class1	30	7	13
Class2	9	18	23
Class3	11	19	20

Table 7:Confusion Matrix

	Class1	Class2	Class3
Precision	0.6	0.4090	0.3571
Recall	0.6	0.36	0.4
F-Measure	0.6	0.3829	0.3773

Table 8:Analysis

Accuracy	45.33%
Mean Precision	0.4554
Mean Recall	0.4533
Mean F-Measure	0.4534

Table 9:Result

➤ **Misclassified Images:**

True: botanical_garden, Predicted: elevator_shaft



Figure 30 Misclassified image as elevator shaft

True: botanical_garden, Predicted: bus_interior



Figure 29 Misclassified image as bus interior

4. Confusion Matrix and Analysis: When L = 1, M = 8

	Actual		
Predicted	Class1	Class2	Class3
Class1	23	13	14
Class2	6	25	19
Class3	10	21	19

Table 230:Confusion Matrix

	Class1	Class2	Class3
Precision	0.6154	0.6122	0.7222
Recall	0.8000	0.6000	0.5200
F-Measure	0.6957	0.6061	0.6047

Table 11:Analysis

Accuracy	44.66%
Mean Precision	0.4596
Mean Recall	0.4466
Mean F-Measure	0.4493

Table 12:Result

5. Confusion Matrix and Analysis: When L = 2, M = 1

	Actual		
Predicted	Class1	Class2	Class3
Class1	40	7	3
Class2	6	34	10
Class3	4	23	23

Table 243:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8	0.5312	0.6388
Recall	0.8	0.68	0.46
F-Measure	0.8	0.5964	0.5348

Table 14:Analysis

Accuracy	64.66%
Mean Precision	0.6567
Mean Recall	0.6466
Mean F-Measure	0.6437

Table 15:Result

6. Confusion Matrix and Analysis: When L = 2, M = 2

	Actual		
Predicted	Class1	Class2	Class3
Class1	33	10	7
Class2	5	33	12
Class3	3	24	23

Table 256:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8048	0.4925	0.5467
Recall	0.66	0.66	0.46
F-Measure	0.7252	0.5641	0.5

Table 17:Analysis

Accuracy	59.33%
Mean Precision	0.6150
Mean Recall	0.5933
Mean F-Measure	0.5964

Table 18:Result

7. Confusion Matrix and Analysis: When L = 2, M = 4

	Actual		
Predicted	Class1	Class2	Class3
Class1	31	12	7
Class2	5	33	12
Class3	4	22	24

Table 19:Confusion Matrix

	Class1	Class2	Class3
Precision	0.775	0.4925	0.5581
Recall	0.62	0.66	0.48
F-Measure	0.6889	0.5641	0.5161

Table 20:Analysis

Accuracy	58.67%
Mean Precision	0.6085
Mean Recall	0.5867
Mean F-Measure	0.5897

Table21:Result

8. Confusion Matrix and Analysis: When L = 2, M = 8

	Actual		
Predicted	Class1	Class2	Class3
Class1	35	9	6
Class2	11	28	11
Class3	7	12	31

Table 22:Confusion Matrix

	Class1	Class2	Class3
Precision	0.6603	0.5714	0.6458
Recall	0.7	0.56	0.62
F-Measure	0.6796	0.5656	0.6326

Table 263:Analysis

Accuracy	62.66%
Mean Precision	0.6258
Mean Recall	0.6266
Mean F-Measure	0.6259

Table 24:Result

9. Confusion Matrix and Analysis: When L = 5, M = 1

	Actual		
Predicted	Class1	Class2	Class3
Class1	43	4	3
Class2	1	40	9
Class3	3	18	29

Table 25:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9148	0.6451	0.7073
Recall	0.86	0.8	0.58
F-Measure	0.8865	0.7142	0.6373

Table 276:Analysis

Accuracy	74.67%
Mean Precision	0.7557
Mean Recall	0.7466
Mean F-Measure	0.7460

Table 27:Result

10. Confusion Matrix and Analysis: When L = 5, M = 2

	Actual		
Predicted	Class1	Class2	Class3
Class1	39	7	4
Class2	6	28	16
Class3	4	20	26

Table 28:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7959	0.5090	0.5652
Recall	0.78	0.56	0.52
F-Measure	0.7878	0.5333	0.5416

Table 289:Analysis

Accuracy	62.00%
Mean Precision	0.6234
Mean Recall	0.62
Mean F-Measure	0.6209

Table 30:Result

11. Confusion Matrix and Analysis: When L = 5, M = 4

	Actual		
Predicted	Class1	Class2	Class3
Class1	37	2	11
Class2	7	22	21
Class3	4	12	34

Table 31:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7708	0.6111	0.5151
Recall	0.74	0.44	0.68
F-Measure	0.7551	0.5116	0.5862

Table 31:Analysis

Accuracy	62.00%
Mean Precision	0.6323
Mean Recall	0.62
Mean F-Measure	0.6176

Table 32:Result

12. Confusion Matrix and Analysis: When L = 5, M = 8

	Actual		
Predicted	Class1	Class2	Class3
Class1	43	3	4
Class2	19	23	8
Class3	22	14	14

Table 33:Confusion Matrix

	Class1	Class2	Class3
Precision	0.5119	0.575	0.5384
Recall	0.86	0.46	0.28
F-Measure	0.6417	0.5111	0.3684

Table 34:Analysis

Accuracy	53.33%
Mean Precision	0.5417
Mean Recall	0.5333
Mean F-Measure	0.5071

Table 295:Result

13. Confusion Matrix and Analysis: When L = 10, M = 1

	Actual		
Predicted	Class1	Class2	Class3
Class1	39	9	2
Class2	6	33	11
Class3	6	17	27

Table 36:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7647	0.5593	0.675
Recall	0.78	0.66	0.54
F-Measure	0.7722	0.6055	0.6

Table 37:Analysis

Accuracy	66.00%
Mean Precision	0.6663
Mean Recall	0.66
Mean F-Measure	0.6592

Table 308:Result

14. Confusion Matrix and Analysis: When L = 10, M = 2

	Actual		
Predicted	Class1	Class2	Class3
Class1	39	10	1
Class2	5	32	13
Class3	7	24	19

Table 39:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7647	0.4848	0.5757
Recall	0.78	0.64	0.38
F-Measure	0.7722	0.5517	0.4578

Table 40:Analysis

Accuracy	60.00%
Mean Precision	0.6048
Mean Recall	0.6
Mean F-Measure	0.5939

Table 41:Result

15. Confusion Matrix and Analysis: When L = 10, M = 4

	Actual		
Predicted	Class1	Class2	Class3
Class1	35	13	2
Class2	15	24	11
Class3	16	19	15

Table 42:Confusion Matrix

	Class1	Class2	Class3
Precision	0.5303	0.4285	0.5357
Recall	0.7	0.48	0.3
F-Measure	0.6034	0.4528	0.3846

Table 43:Analysis

Accuracy	49.33%
Mean Precision	0.4981
Mean Recall	0.4933
Mean F-Measure	0.4802

Table 44:Result

16. Confusion Matrix and Analysis: When L = 10, M = 8

	Actual		
Predicted	Class1	Class2	Class3
Class1	16	6	28
Class2	3	8	39
Class3	1	8	41

Table 45:Confusion Matrix

	Class1	Class2	Class3
Precision	0.6154	0.6122	0.7222
Recall	0.8000	0.6000	0.5200
F-Measure	0.6957	0.6061	0.6047

Table 46:Analysis

Accuracy	43.33%
Mean Precision	0.5144
Mean Recall	0.4333
Mean F-Measure	0.3994

Table 47:Result

17. Confusion Matrix and Analysis: When L = 15, M = 1

	Actual		
Predicted	Class1	Class2	Class3
Class1	43	5	2
Class2	7	31	12
Class3	5	16	29

Table 48:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7818	0.5961	0.6744
Recall	0.86	0.62	0.58
F-Measure	0.8190	0.6078	0.6236

Table 49:Analysis

Accuracy	68.66%
Mean Precision	0.6841
Mean Recall	0.6866
Mean F-Measure	0.6835

Table 50:Result

18. Confusion Matrix and Analysis: When L = 15, M = 2

	Actual		
Predicted	Class1	Class2	Class3
Class1	47	2	1
Class2	9	24	17
Class3	7	15	28

Table 531:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7460	0.5853	0.6086
Recall	0.94	0.48	0.56
F-Measure	0.8318	0.5274	0.5833

Table 5532:Analysis

Accuracy	66.00%
Mean Precision	0.6466
Mean Recall	0.66
Mean F-Measure	0.6475

Table 533:Result

19. Confusion Matrix and Analysis: When L = 15, M = 4

	Actual		
Predicted	Class1	Class2	Class3
Class1	33	6	11
Class2	5	35	10
Class3	1	24	25

Table 54:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8461	0.5384	0.5434
Recall	0.66	0.7	0.5
F-Measure	0.7415	0.6086	0.5208

Table 55:Analysis

Accuracy	62.00%
Mean Precision	0.6426
Mean Recall	0.62
Mean F-Measure	0.6237

Table 56:Result

20. Confusion Matrix and Analysis: When L = 15, M = 8

	Actual		
Predicted	Class1	Class2	Class3
Class1	45	4	1
Class2	20	23	7
Class3	13	10	27

Table 57:Confusion Matrix

	Class1	Class2	Class3
Precision	0.5769	0.6216	0.7714
Recall	0.9	0.46	0.54
F-Measure	0.7031	0.5287	0.6352

Table 58:Analysis

Accuracy	63.33%
Mean Precision	0.6566
Mean Recall	0.6333
Mean F-Measure	0.6223

Table 59:Result

21. Confusion Matrix and Analysis: When L = 20, M = 1

	Actual		
Predicted	Class1	Class2	Class3
Class1	42	5	3
Class2	7	31	12
Class3	5	15	30

Table 60:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7777	0.6078	0.6667
Recall	0.84	0.62	0.6
F-Measure	0.8076	0.6138	0.6315

Table 61:Analysis

Accuracy	68.66%
Mean Precision	0.6840
Mean Recall	0.6866
Mean F-Measure	0.6843

Table 62:Result

22. Confusion Matrix and Analysis: When L = 20, M = 2

	Actual		
Predicted	Class1	Class2	Class3
Class1	42	7	1
Class2	6	27	17
Class3	6	17	27

Table 63:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7778	0.5294	0.6
Recall	0.84	0.54	0.54
F-Measure	0.8076	0.5346	0.5684

Table 64:Analysis

Accuracy	64.00%
Mean Precision	0.6357
Mean Recall	0.64
Mean F-Measure	0.6369

Table 65:Result

23. Confusion Matrix and Analysis: When L = 20, M = 4

	Actual		
Predicted	Class1	Class2	Class3
Class1	11	1	38
Class2	1	4	45
Class3	0	4	46

Table 66:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9166	0.4444	0.3567
Recall	0.22	0.08	0.92
F-Measure	0.3548	0.1355	0.5139

Table 67:Analysis

Accuracy	40.67%
Mean Precision	0.5725
Mean Recall	0.4067
Mean F-Measure	0.3347

Table 68:Result

24. Confusion Matrix and Analysis: When L = 20, M = 8

	Actual		
Predicted	Class1	Class2	Class3
Class1	46	3	1
Class2	12	32	6
Class3	13	23	14

Table 69: Confusion Matrix

	Class1	Class2	Class3
Precision	0.6478	0.5517	0.6667
Recall	0.92	0.64	0.28
F-Measure	0.7603	0.5925	0.3943

Table 70: Analysis

Accuracy	61.33%
Mean Precision	0.6220
Mean Recall	0.6133
Mean F-Measure	0.5824

Table 71: Result

3 Solution Approach

➤ Density estimated from K-nearest neighbour (KNN) method:

The K-Nearest Neighbor (KNN) method for density estimation is a non-parametric technique used to estimate the probability density of a given point in a dataset. Unlike parametric methods, KNN does not assume a specific functional form (e.g., Gaussian) for the data distribution, making it flexible for estimating densities in complex datasets.

Procedure:

- A region with many data points (high density) will have neighbors closer to x , leading to a higher density estimate.
- A region with few data points (low density) will have neighbors farther from x , resulting in a lower density estimate.
- Compute the distance of x from every point in the dataset using a distance metric (e.g., Euclidean distance):

$$d(x, x_i) = \sqrt{\sum_{j=1}^d (x_j - x_{ij})^2}$$

3.1 Results

1. Confusion Matrix and Analysis: when K=1

	Actual		
Predicted	Class1	Class2	Class3
Class1	31	10	9

	Class1	Class2	Class3
Precision	0.7560	0.5084	0.54

Class2	6	30	14
Class3	4	19	27

Table 72:Confusion Matrix

Recall	0.62	0.6	0.54
F-Measure	0.6813	0.5504	0.54

Table73:Analysis

Accuracy	58.67%
Mean Precision	0.6015
Mean Recall	0.5866
Mean F-Measure	0.5905

Table 74:Result

2. Confusion Matrix and Analysis: when K=3

	Actual		
Predicted	Class1	Class2	Class3
Class1	32	8	10
Class2	3	31	16
Class3	1	17	32

Table 75:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8889	0.5535	0.5517
Recall	0.64	0.62	0.64
F-Measure	0.7441	0.5849	0.5925

Table 75:Analysis

Accuracy	63.33%
Mean Precision	0.6647
Mean Recall	0.6333
Mean F-Measure	0.6405

Table 76:Result

3. Confusion Matrix and Analysis: when K=5

	Actual		
Predicted	Class1	Class2	Class3
Class1	33	8	9
Class2	3	31	16
Class3	1	15	34

Table 77:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8918	0.5740	0.5762
Recall	0.66	0.62	0.68
F-Measure	0.7586	0.5961	0.6238

Table 78:Analysis

Accuracy	65.33%
Mean Precision	0.6807
Mean Recall	0.6533
Mean F-Measure	0.6595

Table 79:Result

4. Confusion Matrix and Analysis: when K=7

	Actual		
Predicted	Class1	Class2	Class3
Class1	34	6	10
Class2	3	29	18
Class3	1	15	34

Table 80:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8947	0.58	0.5483
Recall	0.68	0.58	0.68
F-Measure	0.7727	0.58	0.6071

Table81:Analysis

Accuracy	64.66%
Mean Precision	0.6743
Mean Recall	0.6466
Mean F-Measure	0.6732

Table 82:Result

4. Solution Approach

➤ Apply Fisher linear discriminant analysis (FDA)

Fisher Linear Discriminant Analysis (FDA) is a dimensionality reduction technique primarily used for supervised classification problems. It finds a linear combination of features that maximizes the separation between multiple classes by projecting high-dimensional data into a lower-dimensional space (often 1D or 2D).

Procedure:

1. A dataset with N samples, each represented as a d dimensional vector $\mathbf{x}$, along with class labels.
2. Compute the within class scatter matrix $\mathbf{S}_W$ and Compute the between – class scatter matrix $\mathbf{S}_B$.
3. Solve the generalized eigenvalue problem:

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w}$$

4. Select the eigenvectors corresponding to the *largest* $c - 1$ eigenvalues.
5. Project the original data $\mathbf{x}$ onto the reduced space using the projection matrix $\mathbf{W}$ formed by the selected eigenvectors:

$$\mathbf{x}' = \mathbf{W}^T \mathbf{x}$$

4.1 Results

1. bayes Classifier using Unimodal Gaussian

	Actual		
Predicted	Class1	Class2	Class3
Class1	41	4	5
Class2	2	35	13
Class3	1	19	30

Table 83:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9318	0.6034	0.625
Recall	0.82	0.7	0.6
F-Measure	0.8723	0.6481	0.6122

Table 84:Analysis

Accuracy	70.66%
Mean Precision	0.7200
Mean Recall	0.7066
Mean F-Measure	0.7109

Table 85:Result

2. bayes Classifier using Gaussian mixer model (GMM)

	Actual		
Predicted	Class1	Class2	Class3
Class1	38	5	7
Class2	2	33	15
Class3	1	17	32

Table 86:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9268	0.6	0.5925
Recall	0.76	0.66	0.64
F-Measure	0.8351	0.6285	0.6153

Table 87:Analysis

Accuracy	68.66%
Mean Precision	0.7064
Mean Recall	0.6866
Mean F-Measure	0.6930

Table 88:Result

5. Solution Approach

➤ Perceptron Based Classifier

The **Perceptron-based classifier** is one of the simplest types of artificial neural networks. It is primarily used for binary classification and operates by finding a linear decision boundary that separates two classes. When extended to multi-class classification, strategies like **One-vs-Rest** or **One-vs-One** are used.

Procedure:

To extend the Perceptron to multi-class classification, the **One-vs-Rest (OvR)** approach is commonly used:

1. Train a separate binary Perceptron for each class.
2. Treat the current class as the positive class and all other classes as the negative class.
3. Initialize one weight $vector \mathbf{w}_i$ for each class i , where $i = 1, \dots, c$ (for class classification).
4. Initialize the corresponding bias terms b_i to 0.

5.1 Results

1. Multi-Class Perceptron Classifier

	Actual		
Predicted	Class1	Class2	Class3
Class1	22	3	25
Class2	1	20	29
Class3	0	7	43

Table 89:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9556	0.6667	0.4432
Recall	0.44	0.4	0.86
F-Measure	0.6027	0.5	0.5850

Table 90:Analysis

Accuracy	56.67%
Mean Precision	0.6888
Mean Recall	0.5667
Mean F-Measure	0.7562

Table 91:Result

6. Solution Approach

➤ Logistic Regression Classifier

Logistic regression is a supervised learning algorithm primarily used for binary classification, although it can be extended to multi-class classification. It models the probability that a given input x belongs to a particular class using a logistic function (sigmoid).

Procedure:

1. A dataset with input feature vectors x binary labels $y \in \{0, 1\}$
2. Initialize the weight vector w and bias b to zeros or small random values.
3. For each sample x_i compute the linear combination z and apply the sigmoid function:

$$z_i = w \cdot x_i + b, \quad P(y = 1|x_i) = \sigma(z_i)$$

4. Use the **log-loss** (negative log-likelihood) as the cost function:

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(P(y = 1|x_i)) + (1 - y_i) \log(1 - P(y = 1|x_i))]$$

5. Compute gradients of the loss with respect to w and b :

$$\frac{\partial L}{\partial w} = -\frac{1}{n} \sum_{i=1}^n (y_i - P(y = 1|x_i)) x_i$$

$$\frac{\partial L}{\partial b} = -\frac{1}{n} \sum_{i=1}^n (y_i - P(y = 1|x_i))$$

6. Update parameters:

$$w \leftarrow w - \eta \cdot \frac{\partial L}{\partial w}$$

$$b \leftarrow b - \eta \cdot \frac{\partial L}{\partial b}$$

6.1 Results

1. Multi-Class Logistic Regression Classifier

	Actual		
Predicted	Class1	Class2	Class3
Class1	39	6	5
Class2	2	35	13
Class3	2	19	29

Table 92:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9069	0.5833	0.6170
Recall	0.78	0.7	0.58
F-Measure	0.8387	0.6363	0.5979

Table 93:Analysis

Accuracy	68.67%
Mean Precision	0.7024
Mean Recall	0.6866
Mean F-Measure	0.6910

Table 94:Result

7. Solution Approach

➤ SVM based Classifier

Support Vector Machines (SVM) are supervised learning algorithms used for classification and regression. They aim to find the optimal hyperplane that separates data points of different classes with the **maximum margin**. SVMs are versatile due to the use of kernel functions, enabling both linear and non-linear decision boundaries.

Procedure:

1. The goal is to maximize the margin between the hyperplane and the nearest data points (support vectors).
2. Minimize:

$$\frac{1}{2} \|w\|^2$$

3. Subject to constraints:

$$y_i(w \cdot x_i + b) \geq 1, \quad \forall i$$

4. Transform input features into a higher-dimensional space where a linear hyperplane can separate the data.
5. **Linear Kernel**: Suitable for linearly separable data.

$$K(x_i, x_j) = x_i \cdot x_j$$

6. **Polynomial Kernel**: Maps data into a higher polynomial feature space.

$$K(x_i, x_j) = (x_i \cdot x_j + r)^d$$

7. **Gaussian/RBF Kernel**: Useful for non-linear separable data.

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)$$

7.1 Results

1. Linear Kernel: when C=0.1

	Actual		
Predicted	Class1	Class2	Class3
Class1	27	20	3
Class2	3	38	9
Class3	3	26	21

Table 95: Confusion Matrix

	Class1	Class2	Class3
Precision	0.8181	0.4523	0.6363
Recall	0.54	0.76	0.42
F-Measure	0.6506	0.5671	0.5060

Table 96: Analysis

Accuracy	57.33%
Mean Precision	0.6356
Mean Recall	0.5733
Mean F-Measure	0.5745

Table 97: Result

2. Linear Kernel: when C=1

	Actual		
Predicted	Class1	Class2	Class3
Class1	31	10	9
Class2	1	37	12
Class3	1	22	27

Table 98:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9393	0.5362	0.5625
Recall	0.62	0.74	0.54
F-Measure	0.7469	0.6218	0.5510

Table 99:Analysis

Accuracy	63.33%
Mean Precision	0.6793
Mean Recall	0.6333
Mean F-Measure	0.6399

Table 100:Result

3. Linear Kernel: when C=10

	Actual		
Predicted	Class1	Class2	Class3
Class1	38	5	7
Class2	2	37	11
Class3	1	18	31

Table 101:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9268	0.6166	0.6326
Recall	0.76	0.74	0.62
F-Measure	0.8351	0.6727	0.6262

Table 102:Analysis

Accuracy	70.67%
Mean Precision	0.7253
Mean Recall	0.7066
Mean F-Measure	0.7113

Table 103:Result

4. Polynomial Kernel: when C=0.1, Degree =2

	Actual		
Predicted	Class1	Class2	Class3
Class1	33	12	5
Class2	1	42	7
Class3	1	25	24

Table 104:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9428	0.5316	0.6667
Recall	0.66	0.84	0.48
F-Measure	0.7764	0.6511	0.5581

Table 103:Analysis

Accuracy	66.00%
Mean Precision	0.7137
Mean Recall	0.66
Mean F-Measure	0.6619

Table 104:Result

5. Polynomial Kernel: when C=0.1, Degree =3

	Actual		
Predicted	Class1	Class2	Class3
Class1	38	7	5
Class2	3	40	7
Class3	1	21	28

Table 3405:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9047	0.5882	0.7067
Recall	0.76	0.8	0.56
F-Measure	0.8260	0.6779	0.6222

Table 106:Analysis

Accuracy	70.67%
Mean Precision	0.7309
Mean Recall	0.7066
Mean F-Measure	0.7087

Table 107:Result

6. Polynomial Kernel: when C=0.1, Degree =4

	Actual		
Predicted	Class1	Class2	Class3
Class1	41	5	4
Class2	5	38	7
Class3	1	20	29

Table 3508:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8723	0.6031	0.725
Recall	0.82	0.76	0.58
F-Measure	0.8453	0.6725	0.6444

Table 109:Analysis

Accuracy	72.00%
Mean Precision	0.7337
Mean Recall	0.7200
Mean F-Measure	0.7207

Table 110:Result

7. Polynomial Kernel: when C=1, Degree =2

	Actual		
Predicted	Class1	Class2	Class3
Class1	41	5	4
Class2	4	38	8
Class3	1	18	31

Table 1136:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8913	0.6229	0.7209
Recall	0.82	0.76	0.62
F-Measure	0.8541	0.6846	0.6667

Table 1137:Analysis

Accuracy	73.33%
Mean Precision	0.7450
Mean Recall	0.7333
Mean F-Measure	0.7351

Table 1138:Result

4. Polynomial Kernel: when C=1, Degree =3

	Actual		
Predicted	Class1	Class2	Class3
Class1	43	3	4
Class2	6	37	7
Class3	2	18	30

Table 3914:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8431	0.6379	0.7317
Recall	0.86	0.74	0.6
F-Measure	0.8514	0.6851	0.6593

Table 115:Analysis

Accuracy	73.33%
Mean Precision	0.7375
Mean Recall	0.7333
Mean F-Measure	0.7320

Table 116:Result

4. Polynomial Kernel: when C=1, Degree =4

	Actual		
Predicted	Class1	Class2	Class3
Class1	41	4	5
Class2	5	38	7
Class3	4	17	29

Table 4017:Confusion Matrix

	Class1	Class2	Class3
Precision	0.82	0.6440	0.7073
Recall	0.82	0.76	0.58
F-Measure	0.82	0.6972	0.6373

Table 118:Analysis

Accuracy	72.00%
Mean Precision	0.7237
Mean Recall	0.7200
Mean F-Measure	0.7182

Table 119:Result

4. Polynomial Kernel: when C=10, Degree =2

	Actual		
Predicted	Class1	Class2	Class3
Class1	41	4	5
Class2	5	35	10
Class3	2	18	30

Table 4120:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8541	0.6140	0.6667
Recall	0.82	0.7	0.6
F-Measure	0.8367	0.6542	0.6315

Table 121:Analysis

Accuracy	70.67%
Mean Precision	0.7116
Mean Recall	0.7067
Mean F-Measure	0.7075

Table 1242:Result

4. Polynomial Kernel: when C=10, Degree =3

	Actual		
Predicted	Class1	Class2	Class3

	Class1	Class2	Class3
--	--------	--------	--------

Class1	40	4	6
Class2	7	37	6
Class3	4	19	27

Table 4324:Confusion Matrix

Precision	0.7843	0.6166	0.6923
Recall	0.8	0.74	0.54
F-Measure	0.7920	0.6727	0.6067

Table 1445:Analysis

Accuracy	69.33%
Mean Precision	0.6977
Mean Recall	0.6933
Mean F-Measure	0.6905

Table 126:Result

4. Polynomial Kernel: when C=10, Degree =4

	Actual		
Predicted	Class1	Class2	Class3
Class1	39	5	6
Class2	7	35	8
Class3	5	12	33

Table 4527:Confusion Matrix

	Class1	Class2	Class3
Precision	0.7647	0.6730	0.7021
Recall	0.78	0.7	0.66
F-Measure	0.7722	0.6862	0.6804

Table 1468:Analysis

Accuracy	71.33%
Mean Precision	0.7133
Mean Recall	0.7133
Mean F-Measure	0.7129

Table 129:Result

4. Gaussian/RBF Kernel: when C=0.1, Gamma =0.01

	Actual		
Predicted	Class1	Class2	Class3
Class1	27	20	3
Class2	3	38	9
Class3	3	26	21

Table 130:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8181	0.4523	0.6363
Recall	0.54	0.76	0.42
F-Measure	0.6506	0.5671	0.5060

Table 131:Analysis

Accuracy	57.33%
Mean Precision	0.6356
Mean Recall	0.5733
Mean F-Measure	0.5745

Table 1472:Result

4. Gaussian/RBF Kernel: when C=0.1, Gamma =0.1

	Actual		
Predicted	Class1	Class2	Class3
Class1	27	20	3
Class2	2	40	8
Class3	3	26	21

Table 4833:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8437	0.4651	0.6562
Recall	0.54	0.80	0.42
F-Measure	0.6585	0.5882	0.5121

Table 134:Analysis

Accuracy	58.67%
Mean Precision	0.6550
Mean Recall	0.5866
Mean F-Measure	0.5863

Table 135:Result

4. Gaussian/RBF Kernel: when C=0.1, Gamma =1

	Actual		
Predicted	Class1	Class2	Class3
Class1	25	22	3
Class2	2	43	5
Class3	2	28	20

Table 4936:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8620	0.4623	0.7142
Recall	0.5	0.86	0.4
F-Measure	0.6329	0.6013	0.5128

Table 137:Analysis

Accuracy	58.67%
Mean Precision	0.6795
Mean Recall	0.5867
Mean F-Measure	0.5823

Table 138:Result

4. Gaussian/RBF Kernel: when C=1, Gamma =0.01

	Actual		
Predicted	Class1	Class2	Class3
Class1	27	20	3
Class2	3	38	9
Class3	3	26	21

Table 5039:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8181	0.4523	0.6363
Recall	0.54	0.76	0.42
F-Measure	0.6506	0.5671	0.5060

Table 140:Analysis

Accuracy	57.33%
Mean Precision	0.6356
Mean Recall	0.5733
Mean F-Measure	0.5745

Table 141:Result

4. Gaussian/RBF Kernel: when C=1, Gamma =0.1

	Actual		
Predicted	Class1	Class2	Class3
Class1	27	20	3
Class2	2	40	8
Class3	3	26	21

Table 5142:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8437	0.4651	0.6562
Recall	0.54	0.80	0.42
F-Measure	0.6585	0.5882	0.5121

Table 143:Analysis

Accuracy	58.67%
Mean Precision	0.6550

Mean Recall	0.5866
Mean F-Measure	0.5863

Table 144:Result

4. Gaussian/RBF Kernel: when C=1, Gamma =1

	Actual		
Predicted	Class1	Class2	Class3
Class1	36	10	4
Class2	1	42	7
Class3	1	22	27

Table 5245:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9437	0.5675	0.7105
Recall	0.72	0.84	0.54
F-Measure	0.8181	0.6774	0.6136

Table 146:Analysis

Accuracy	70.00%
Mean Precision	0.7418
Mean Recall	0.7000
Mean F-Measure	0.7030

Table 147:Result

4. Gaussian/RBF Kernel: when C=10, Gamma =0.01

	Actual		
Predicted	Class1	Class2	Class3
Class1	27	20	3
Class2	3	38	9
Class3	3	26	21

Table148:Confusion Matrix

	Class1	Class2	Class3
Precision	0.8181	0.4523	0.6363
Recall	0.54	0.76	0.42
F-Measure	0.6506	0.5671	0.5060

Table 149:Analysis

Accuracy	57.33%
Mean Precision	0.6356
Mean Recall	0.5733
Mean F-Measure	0.5745

Table 150:Result

4. Gaussian/RBF Kernel: when C=10, Gamma =0.1

	Actual		
Predicted	Class1	Class2	Class3
Class1	35	6	9
Class2	2	37	11
Class3	1	22	27

Table 151:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9210	0.5691	0.5744
Recall	0.7	0.74	0.54
F-Measure	0.7954	0.6434	0.5567

Table 152:Analysis

Accuracy	66.00%
Mean Precision	0.6882
Mean Recall	0.66
Mean F-Measure	0.6652

Table 153:Result

4. Gaussian/RBF Kernel: when C=10, Gamma =1

	Actual		
Predicted	Class1	Class2	Class3
Class1	41	5	4
Class2	3	42	8
Class3	1	18	31

Table 154:Confusion Matrix

	Class1	Class2	Class3
Precision	0.9111	0.6290	0.7209
Recall	0.70	0.78	0.62
F-Measure	0.7954	0.6434	0.5567

Table 155:Analysis

Accuracy	74.00%
Mean Precision	0.7536
Mean Recall	0.7400
Mean F-Measure	0.7420

Table 156:Result

C. Dimensionality Reduction Analysis

1. PCA Analysis

- **Eigenvalue Distribution:**
 - Demonstrated the significance of the first few principal components.
- **2D Representation:**
 - Visualized class separability post-reduction.

2. FDA Analysis

- Improved discrimination between classes in reduced dimensions.

IV. Overall Comparative Analysis

A. Cross-Assignment Comparison

- Observed performance improvements over previous assignments due to advanced techniques.
- SVM and GMM consistently emerged as top-performing methods.

V. Conclusion

A. Key Findings

- SVM with Gaussian kernel and GMM-based classifiers outperformed others on non-linear data.
- Dimensionality reduction via PCA and FDA improved computational efficiency without sacrificing accuracy.

B. Best Performing Methods

- **Dataset 1:** SVM with RBF kernel.
- **Dataset 2:** GMM with PCA-based dimensionality reduction.

C. Analysis Insights

- Parameter tuning and dimensionality reduction are crucial for high-dimensional datasets.
- Non-linear models excel in capturing complex patterns.

VI. References

- **Bishop, C. M. (2006).** *Pattern Recognition and Machine Learning*. Springer.
- **Duda, R. O., Hart, P. E., & Stork, D. G. (2000).** *Pattern Classification*. Wiley.
- **Murphy, K. P. (2012).** *Machine Learning: A Probabilistic Perspective*. MIT Press.
- **Mitchell, T. M. (1997).** *Machine Learning*. McGraw-Hill.